Introduction and Application of GIS

Topic Covered

Introduction to GIS

Introduction to Remote Sensing

Applications of GIS and RS

What is GIS??

Geographic

+

Information

+

System









Information and **Study** of the earth with reference to their **Location**

Images Courtesy: Google

What is GIS??

- ❑ There are a number of definitions of GIS
- ✓ GIS is **much more** than a container of **maps in digital form**.
- ✓ A GIS is a system (hardware + database engine) that is designed to efficiently, assemble, store, update, analyze, manipulate, and display **geographically referenced information** (data identified by their locations).
- ✓ A system that deals with all types of geographically referenced data
- ✓ A **system designed** to capture, store, manipulate, analyze, manage, and present all types of **geographically referenced data**
- ✓ **Maps + Information = GIS**

What is GIS??

❑ GEOGRAPHIC

- ✓ implies **that locations of the data** items are known, or can be calculated, in terms of **Geographic coordinates** (Latitude, Longitude)

❑ INFORMATION

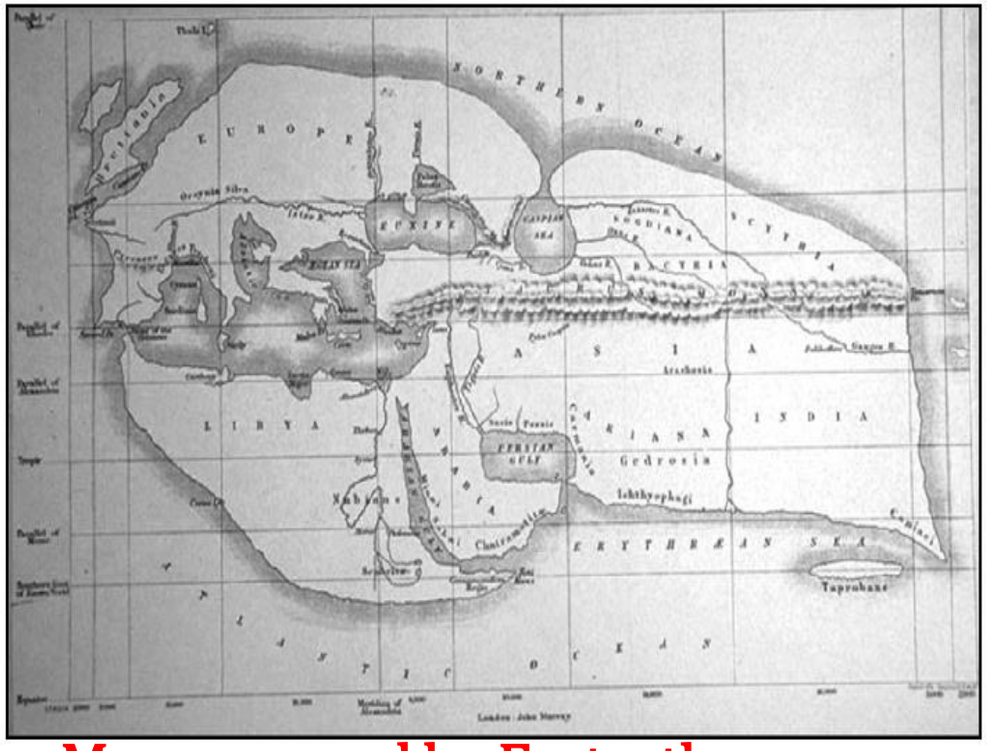
- ✓ implies that the **data** in a GIS are organized **to yield useful knowledge**, often as **colored maps and images**, but also as **statistical graphics, tables**, and various on-screen responses to interactive queries.

❑ SYSTEM

- ✓ implies that a GIS is made up from several **inter-related and linked components with different functions**.
- ✓ Thus, GIS have functional capabilities for **data capture, input, manipulation, transformation, visualization, combinations, query, analysis, modeling and output.**

GIS - History

- Geographical information systems evolved from centuries of mapmaking and the compilation of spatial data



Map prepared by Eratosthenes

(ca. 276 – 194B.C.)



Ptolemy's map of the world, about A.D. 150, republished in 1482. Notice the use of latitude and longitude lines and the distinctive projection of this map

Image Source : DR. MOHSIN SIDDIQUE PPT

GIS - History

- Dr. John Snow and the 1854 Cholera outbreak in London's Broad Street region
- Dr John Snow is known as the ‘father of modern epidemiology’ and the ‘father of GIS’ because of the famous case of the 1854 Cholera outbreak in London’s Broad Street region.



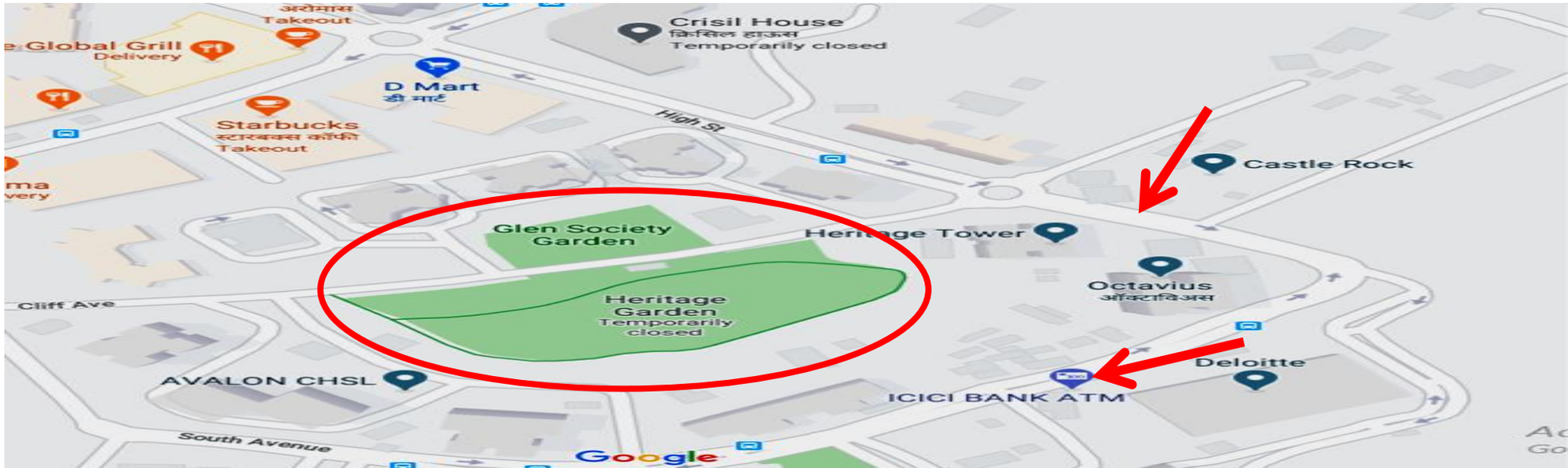




Images Courtesy: Google

What is Map?

- Map is a **graphical representation** of the **features** on the earth's surface on a **two dimensional plane to a scale.**
- In map all objects are represented by **point, line and polygon/area.**
- The process of preparing a map is **mapping**
- The process of **extracting meaningful information** from the map is map reading



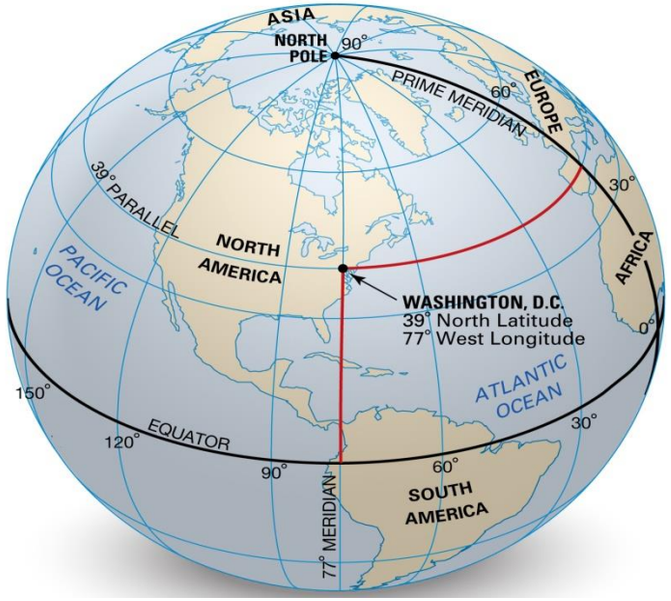
Images Courtesy: Google

Map – Characterizing Geographic Features

- All geographic features on the earth's surface can be characterized as one of three basic feature types:
- **Point data** exists when a feature is associated with a single location in space.
  - ✓ Ex. well, a weather station, etc.
- **Linear data** exists when a feature's location is described by a string of spatial coordinates.
  - ✓ Ex. rivers, roads, pipelines, etc.
- **Areal data** exists when a feature is described by a closed string of spatial coordinates Commonly referred to as a polygon.
  - ✓ Ex. forest stands, soil classification areas, administrative boundaries, and climate zones, etc.

Map – Representation

- This is how we represents 3D earth on flat paper as 2D (Map)
- When this transformation is performed, there are four aspect which needs to be considered to maintain accuracy :
- ✓ Shape, Area, Length / Distance, and Direction





Images Courtesy: Google

Map - Map Projection

- A **map projection** is a method for taking the **curved surface** of the earth and **displaying it on something flat**, like a computer screen or a piece of paper
- **No flat representation** of the earth can be **completely accurate**, and therefore, so many different projections have been developed, each suited to a particular purpose.
- Map projections differ in the way they handle four properties: **Area, Angles/shape, Length/Distance, and Direction**

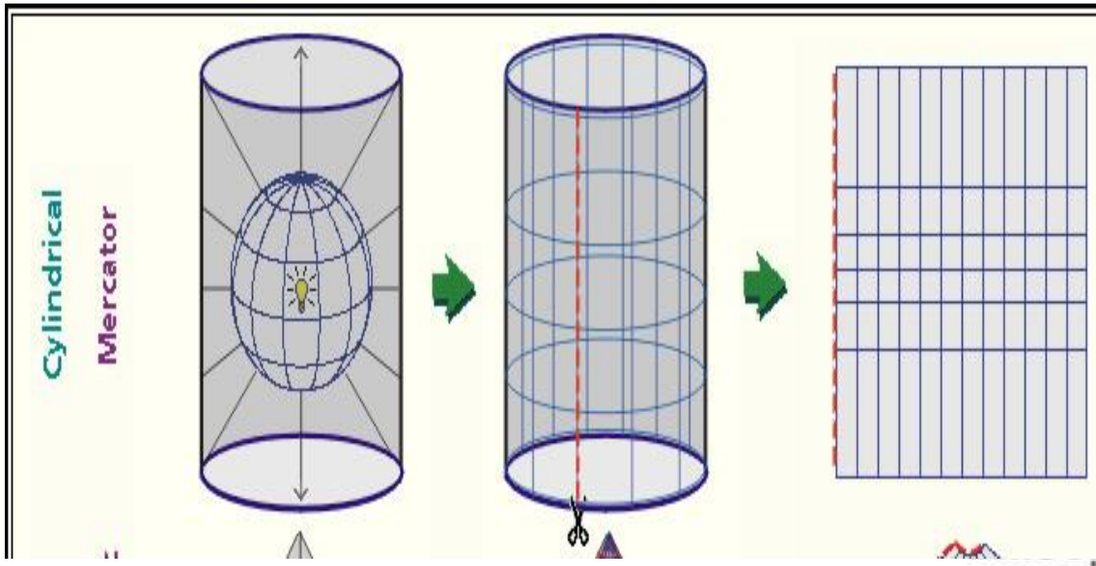
✓ **No projection can preserve all four simultaneously**

✓ **No projection can preserve both Area and Angles,** however. the map-maker must decide which property is most important and choose a projection based on that

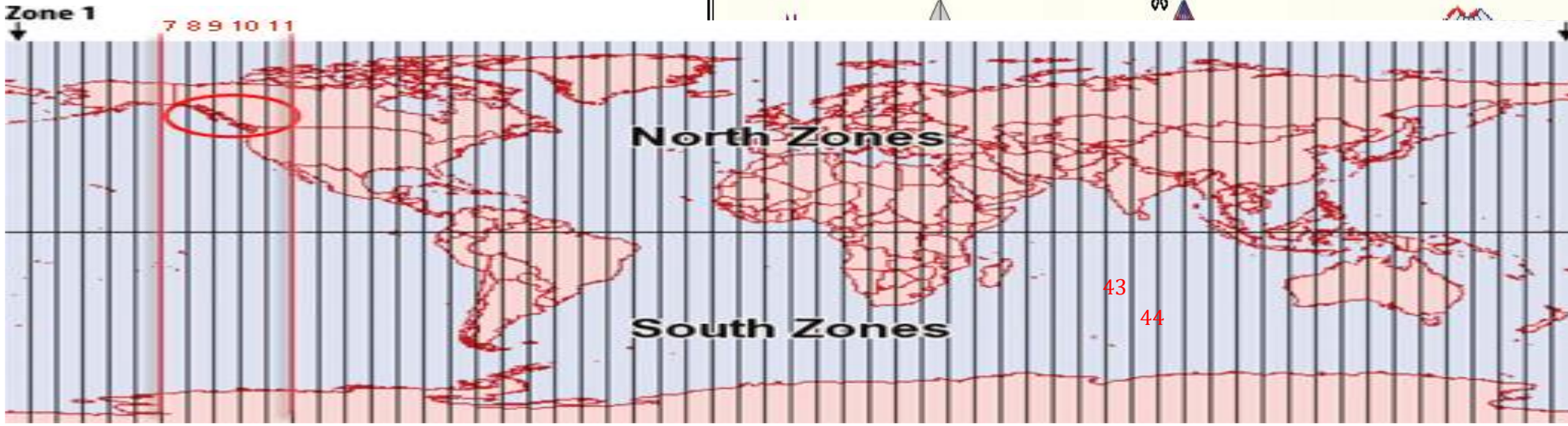
Map - Types of Map Projection

1. Cylindrical Projection

- Preserve distance or/and area
- Easy to use
- Latitude and longitude are right angles
- Distortions are high latitudes



Universal Tranverse Mercator (UTM)
Total Zone = 60
Each zone = 6^0

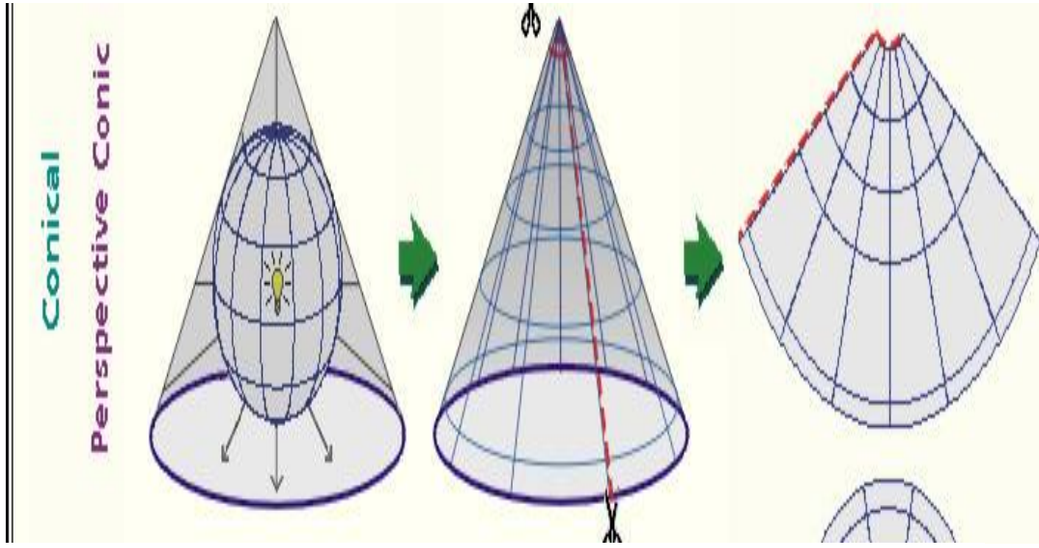


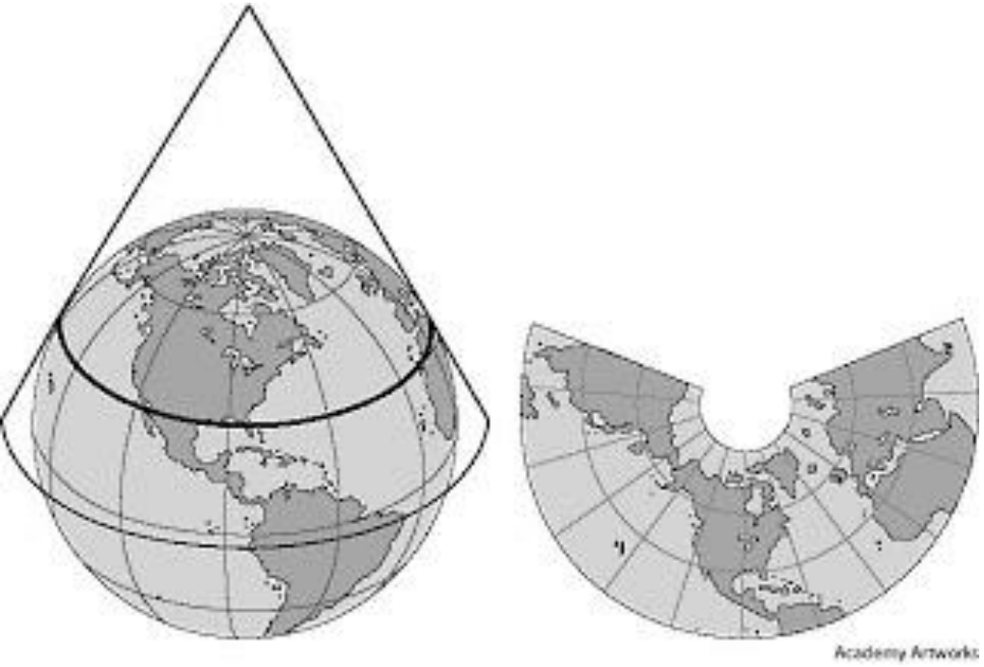
Images Courtesy: Google

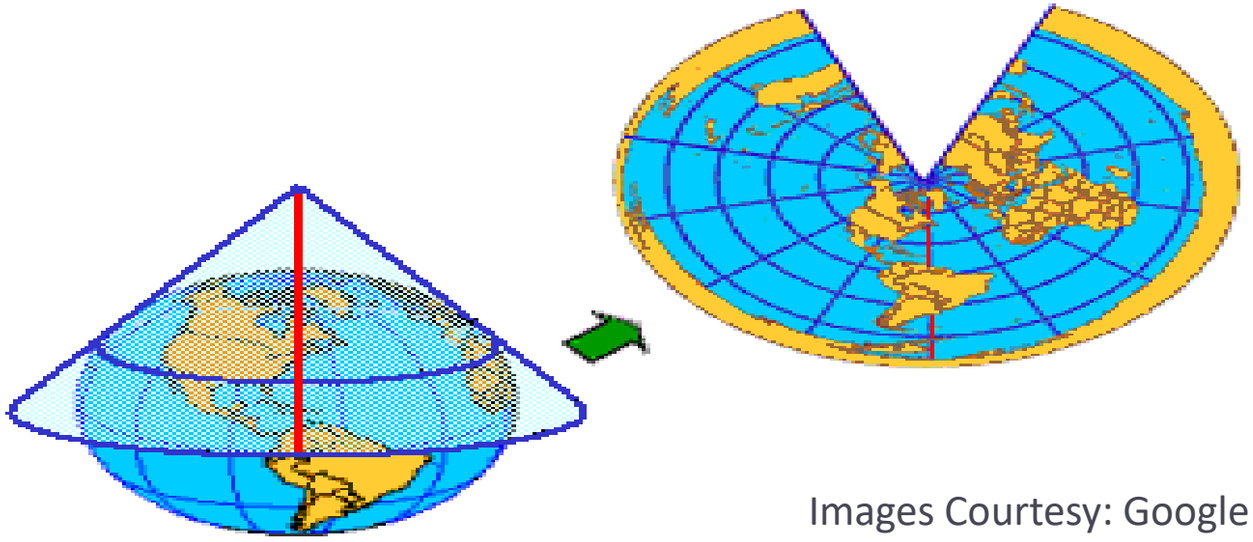
Map - Types of Map Projection

2. Conic projections

- Good for showing small areas accurately
- distortion increases north and south of the standard parallel





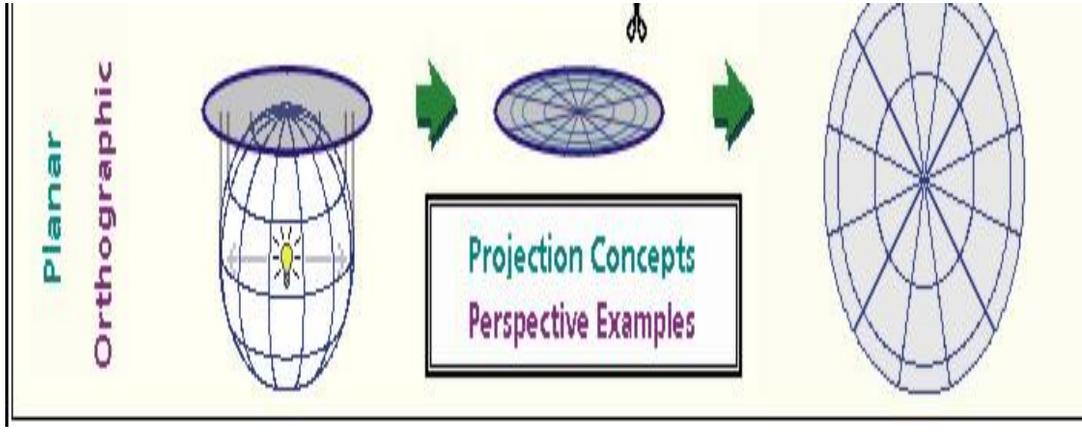


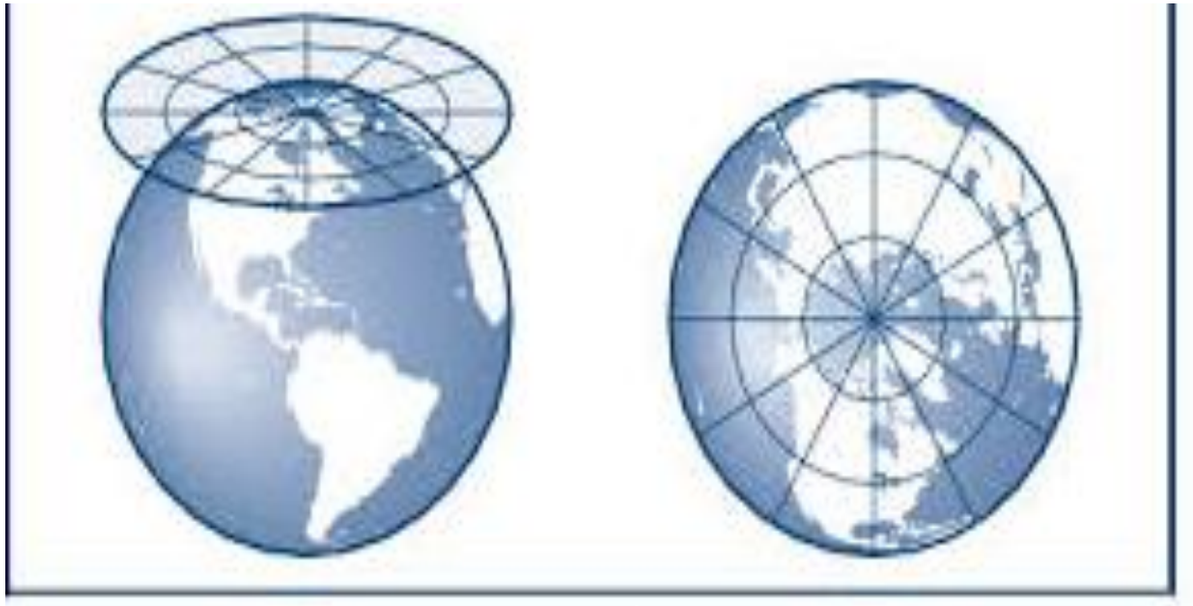
Images Courtesy: Google

Map - Types of Map Projection

3. Planar/Azimuthal projections

- Good for showing hemisphere with with one focus
- Distorts shape and area near edges due to perspective
- Most azimuthal projections are not suitable for displaying the entire Earth in one view, but give a sense of the globe





Images Courtesy: Google

Map - Importance of Map Projection

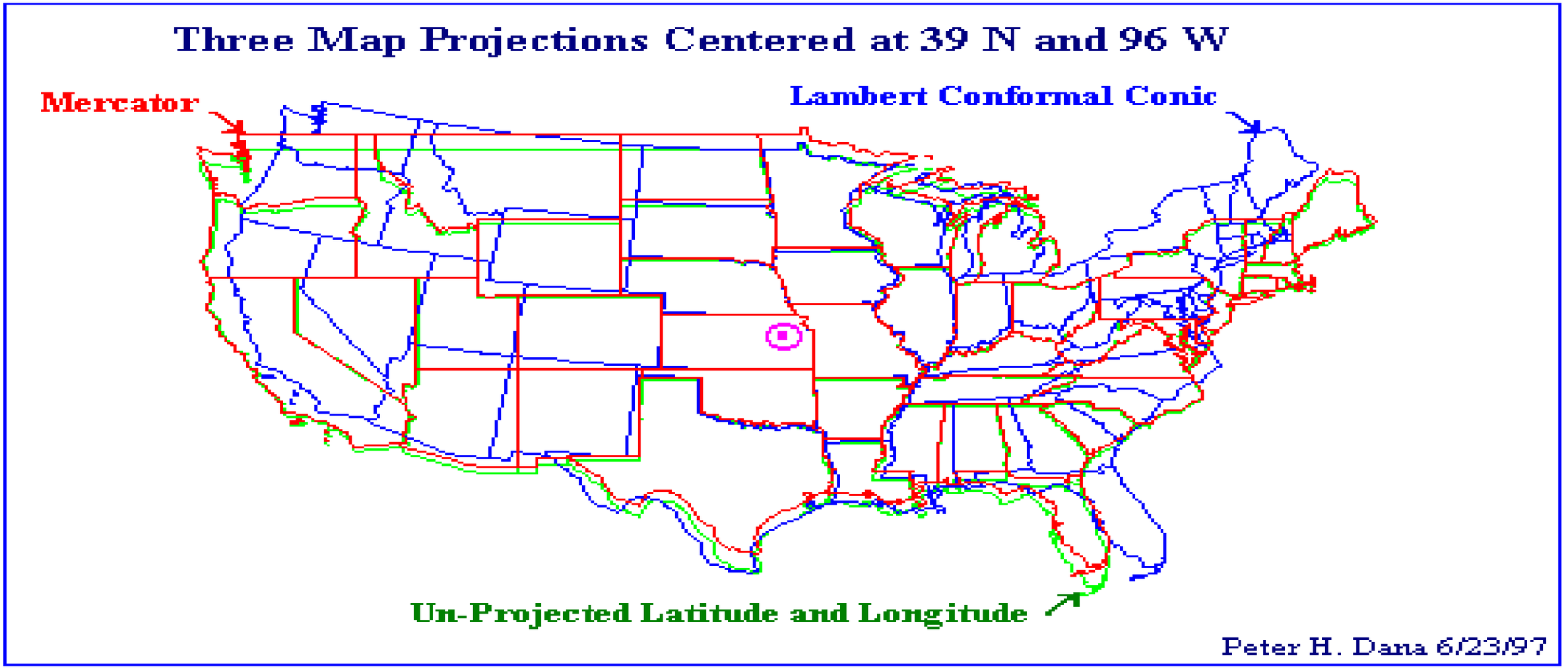
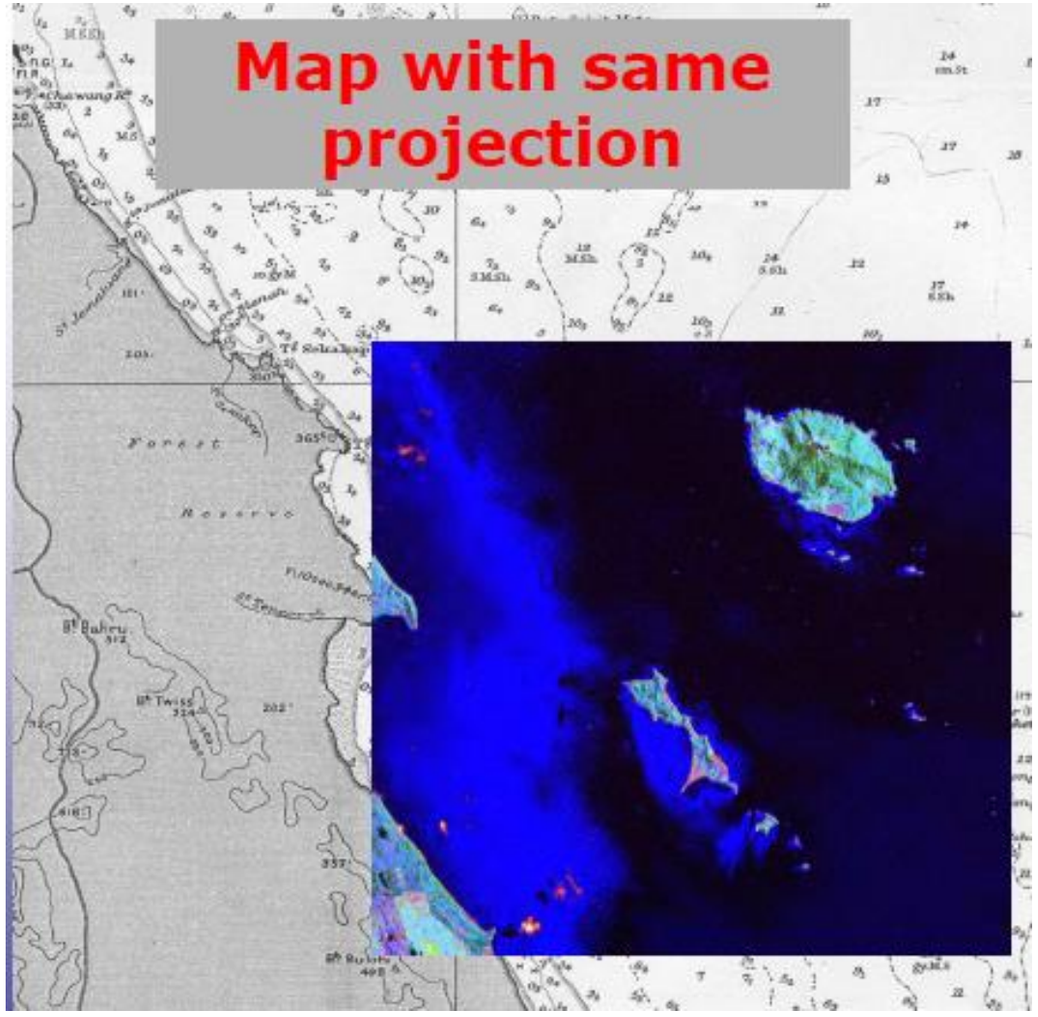
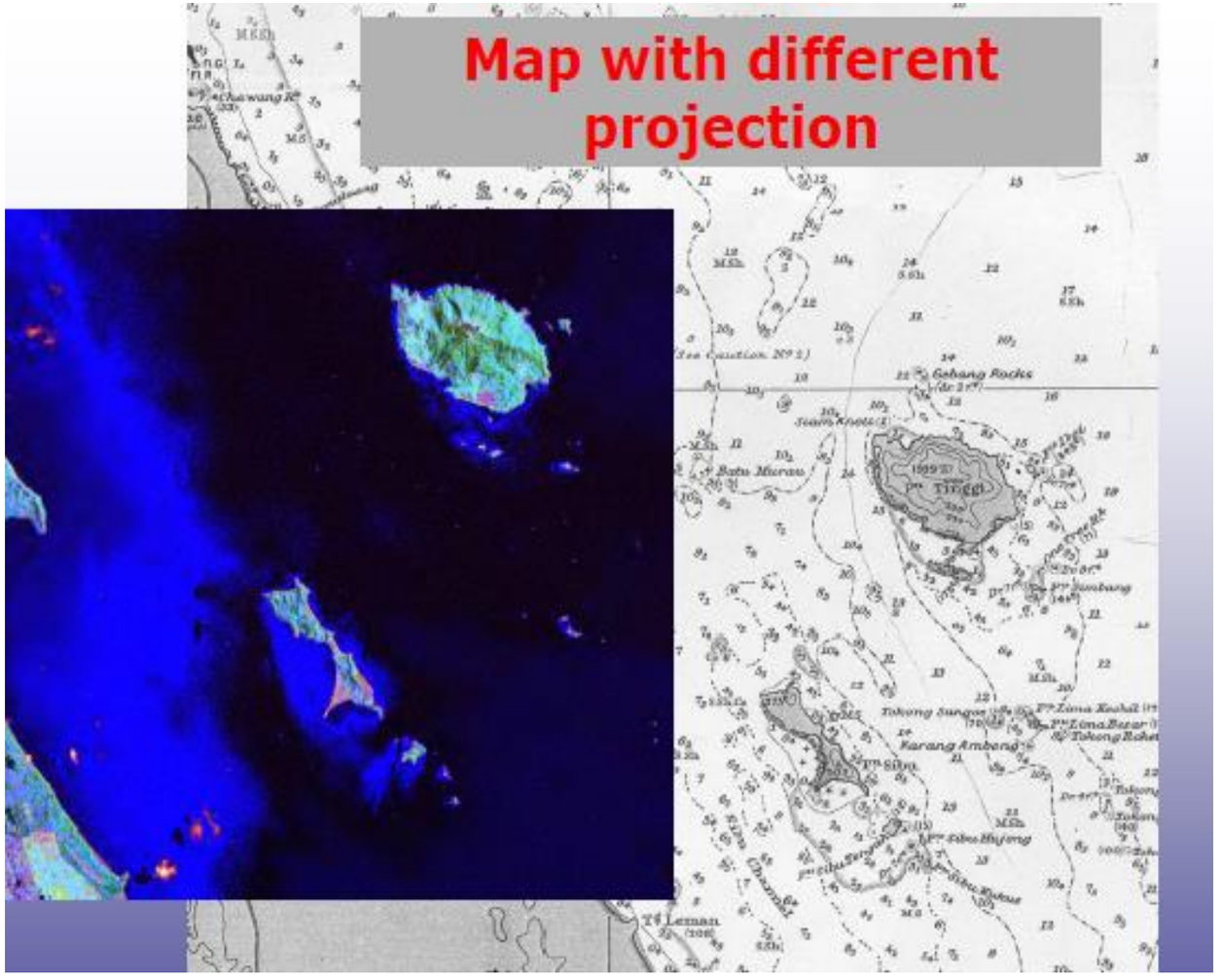


Image courtesy: http://maps.unomaha.edu/Peterson/gis/notes/MapProjCoord.html

Map - Importance of Map Projection





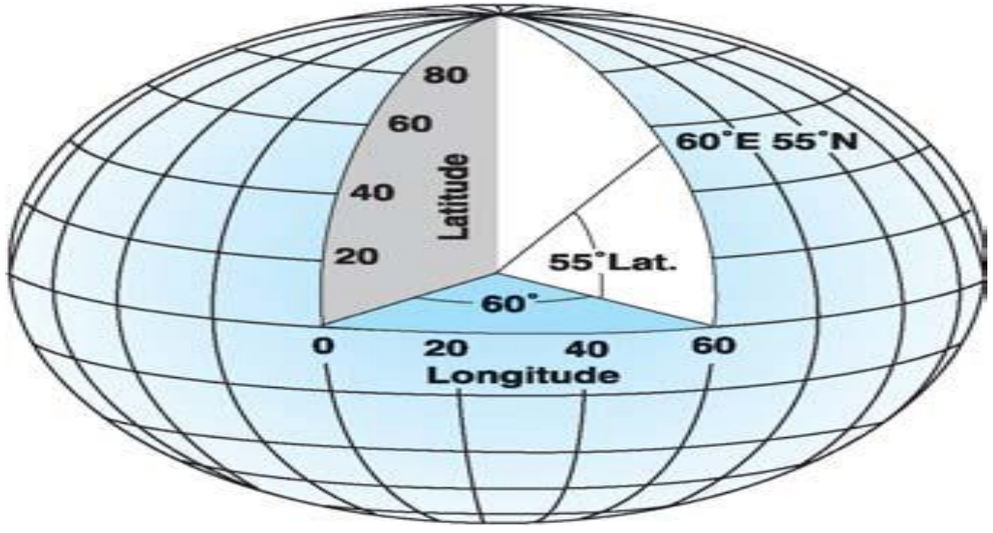
Images Source: Presentation of Dr. V. M. Rokade

GIS - Geographic Co-ordinate Systems (GCS)

- **Latitude and longitude** are ***imaginary (unreal) lines*** drawn on maps to easily locate places on the Earth.
- Latitude is **distance north or south of the equator**
- Longitude is **distance east or west of the prime meridian**
- Both are **measured from the center of earth** in terms of the 360 degrees (symbolized by °) of a circle.
- For Ex. Latitude = 55^0 N

  whereas longitude = 60^0 E

range ,Lat = +90^0 N to -90° S and Long = +180 E to -180 W



categories of spatial referencing systems



- Census geography (postal codes)
- Geographic identifier reference system
- Location specified by a label or code



- Geocoding (postal address)
- Linear reference system
- Location specified by reference to a segment of a linear geographic feature and distance from that segment to a point



- Latitude, Longitude (Degrees, Mins., Secs.)
- Coordinate reference system
- Location specified with respect to a datum

GIS - Digital Maps

❖Easy customizing/ Map Editing/Updation of Informations

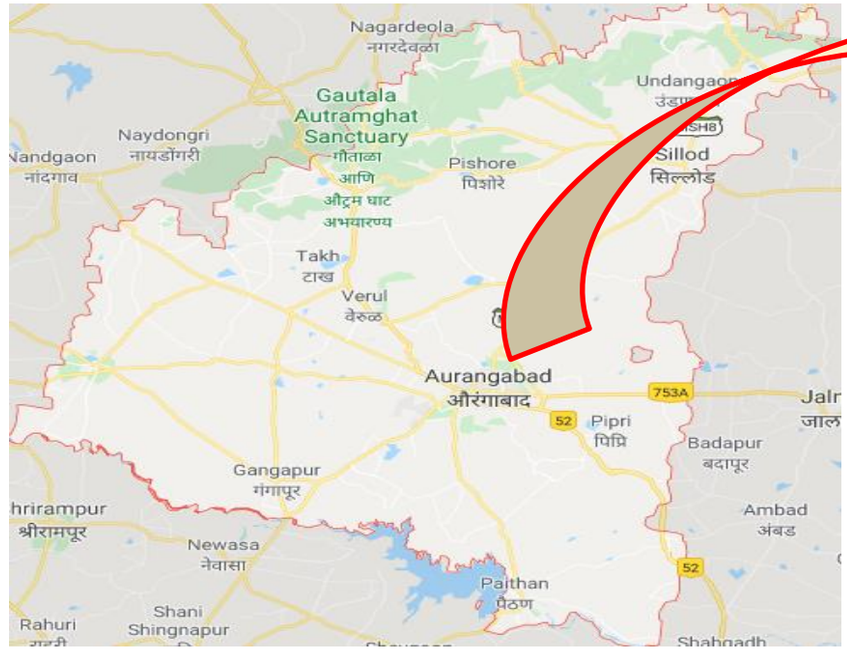
❖Attachment of database attributes to features.

❖Safe storage/ Durability/ huge Storage Capacity/No decomposition/Physical Damage

❖Quick creation of buffers.

❖Overlays and area calculations.





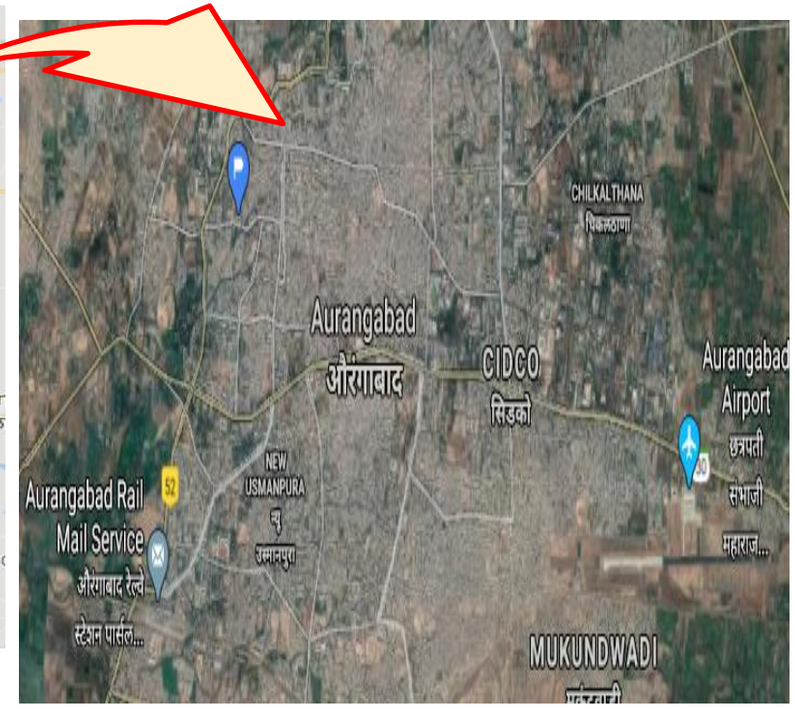
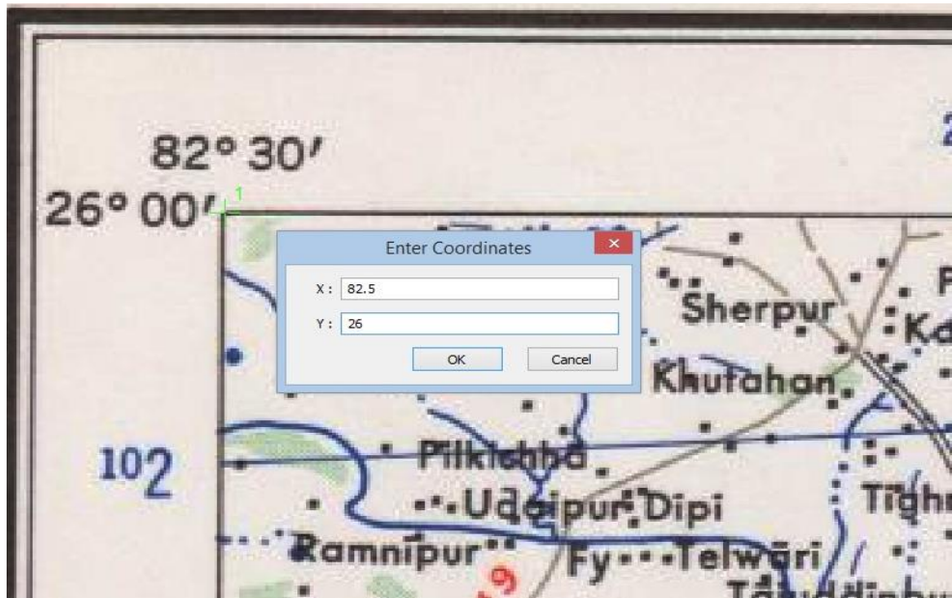


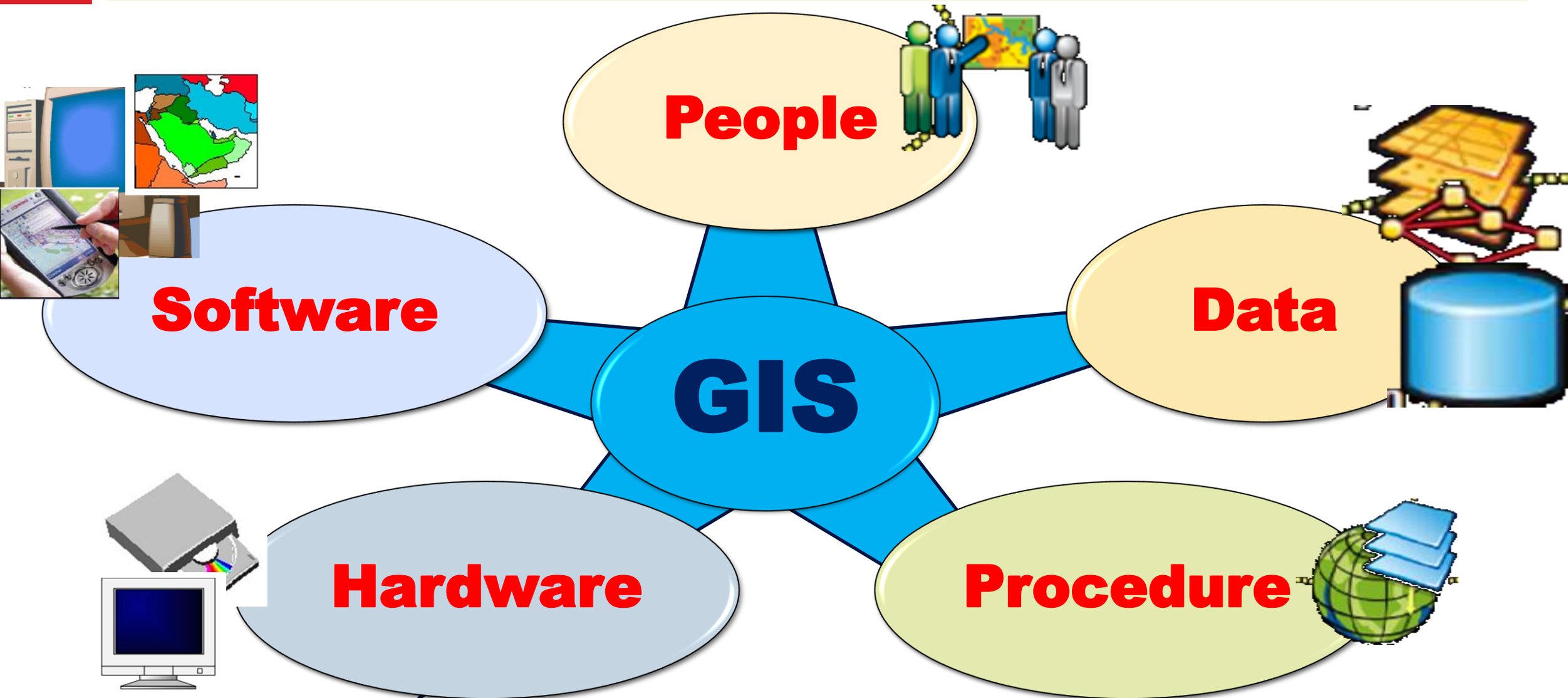
Image source : Maharashtra Remote Sensing Application Centre; Google Maps

GIS - Geographically Referenced Data

- Everything in the digital map is geographically referenced
- Geographically referenced data means data is referenced on earth surface
- It is also called as spatially referenced data
- If data is spatially referenced, then we can use that data in GIS
- Georeferencing transforms images /maps from geometric coordinate system to geographic coordinate system using base map /image (having geographic coordinates).



GIS – Key Component



GIS – Key Component

❑ People

- ✓ GIS users range from technical specialists who design and maintain the system to those who use it to help them perform their everyday work.

❑ Data:-

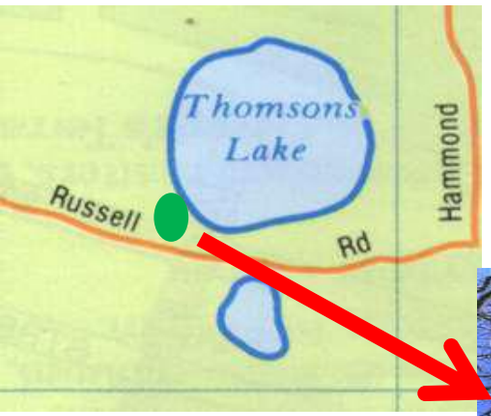
- ✓ GIS incorporates geographical features with tabular data in order to map, analyze, and assess real-world problems.
- ✓ Data that is in some way referenced to locations on the earth.
- ✓ Attribute data can be generally defined as additional information about each of the spatial features.
- ✓ Geographic data and related tabular data can be produced by digitizing images from aerial photographs or published maps.

GIS – Key Components

- **Method/Procedure**
  - ✓ A successful GIS operates according to a well-designed Plan/ procedure
- **Hardware**
  - ✓ Computer
  - ✓ Digitizer and Scanner
  - ✓ Printer/Plotter
- **Software**
  - ✓ Web-based GIS
  - ✓ Geobrowser: Google Earth
  - ✓ Desktop GIS: ArcGIS
  - ✓ QGIS

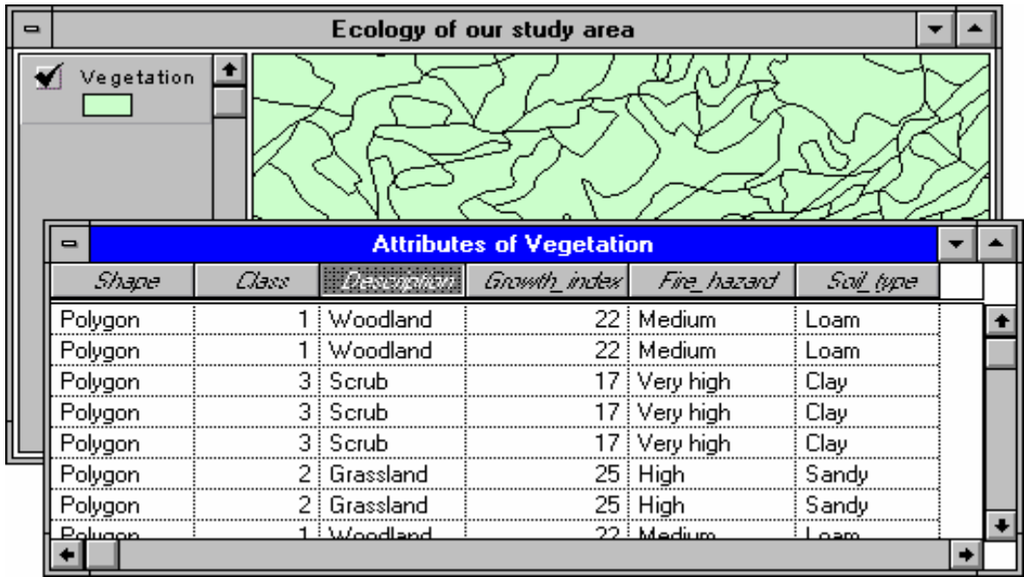
GIS – Data Types

- Accordingly, GIS technology utilizes two basic types of data
- **Spatial Data -** Describes the absolute and relative location of geographic features.
- **Attribute data -** Describes characteristics of the spatial features
  - ✓ A GIS integrates the above two types of data and allows the user to derive new data, for planning and resources management through well defined operations





– Height = 15m
– Diameter = 0.75m
– Species = Oak
– Condition = Good
– Age = 8 years



Shape	Class	Description	Growth_index	Fire_hazard	Soil_type
Polygon	1	Woodland	22	Medium	Loam
Polygon	1	Woodland	22	Medium	Loam
Polygon	3	Scrub	17	Very high	Clay
Polygon	3	Scrub	17	Very high	Clay
Polygon	3	Scrub	17	Very high	Clay
Polygon	2	Grassland	25	High	Sandy
Polygon	2	Grassland	25	High	Sandy
Polygon	1	Woodland	22	Medium	Loam

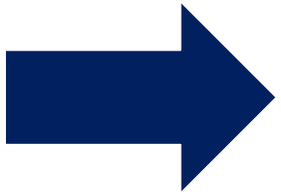
GIS – Spatial Data

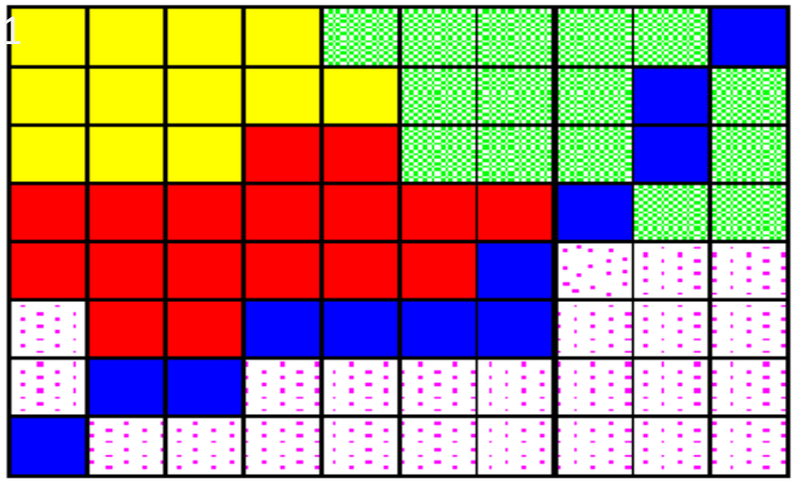
- Traditionally spatial data has been stored and presented in the form of a map.
- Two basic types of spatial data models have evolved for storing geographic data digitally. These are referred to as:
  - ✓ Raster
  - ✓ Vector

GIS – Spatial Data : Raster

- Data is classified as “continuous” (such as in an image), or “thematic” (where each cell denotes a feature type)
- Stores images as rows and columns of numbers with a Digital Value/Number (DN) for each cell
- Units are usually represented as square grid cells that are uniform in size
- Numerous data formats (TIFF, GIF, ERDAS img, JPEG, etc)

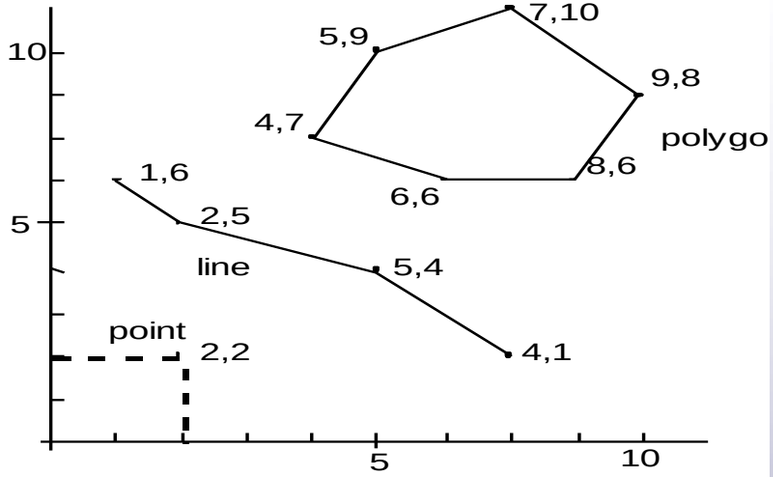


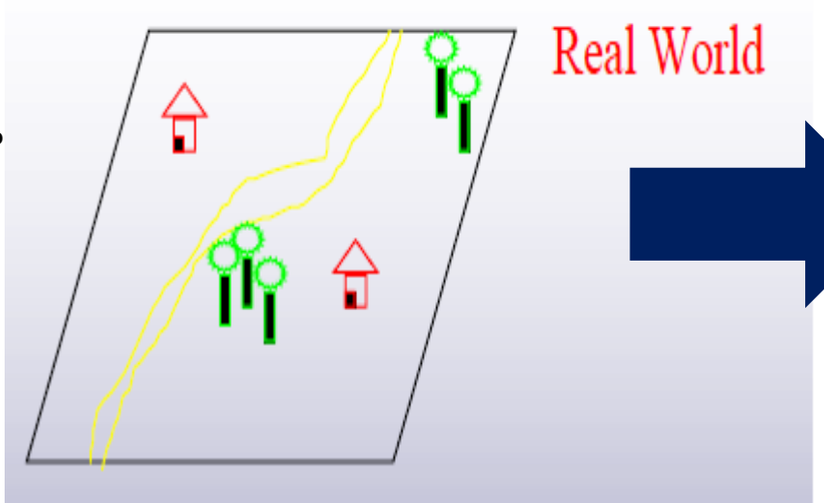


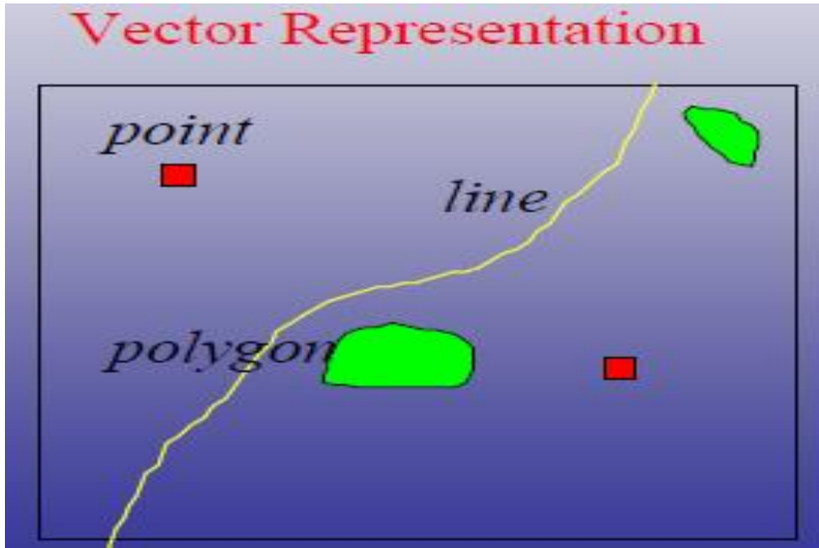


GIS – Spatial Data : Vector

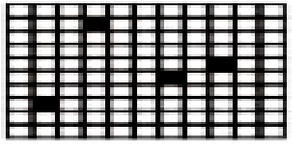
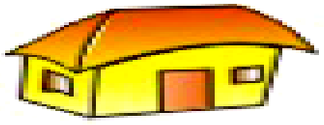
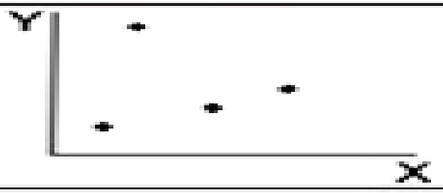
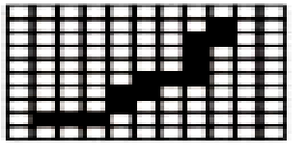
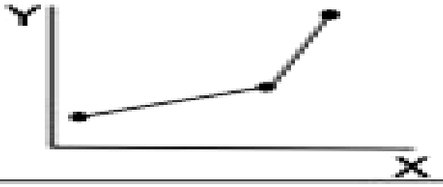
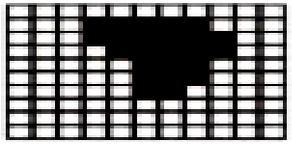
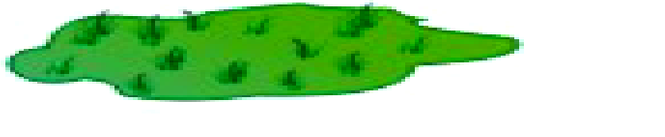
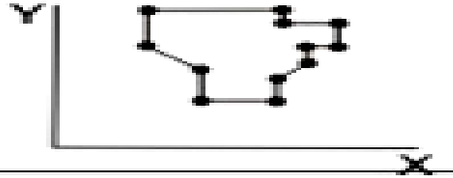
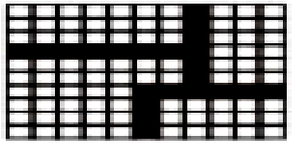
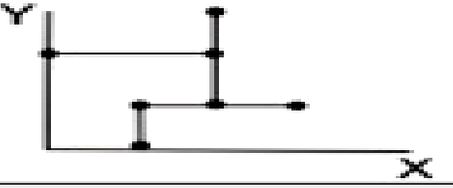
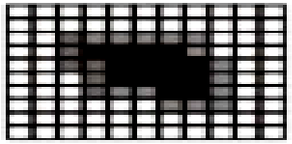
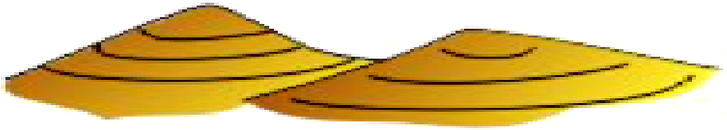
- Allows user to specify specific spatial locations
- It not broken up into discrete grid squares
- Information about points, lines, and polygons is encoded and stored as a collection of x & y coordinates.
- Numerous data formats- SHP, KML, GNT, DWG (AutoCad Map Drawing) etc.





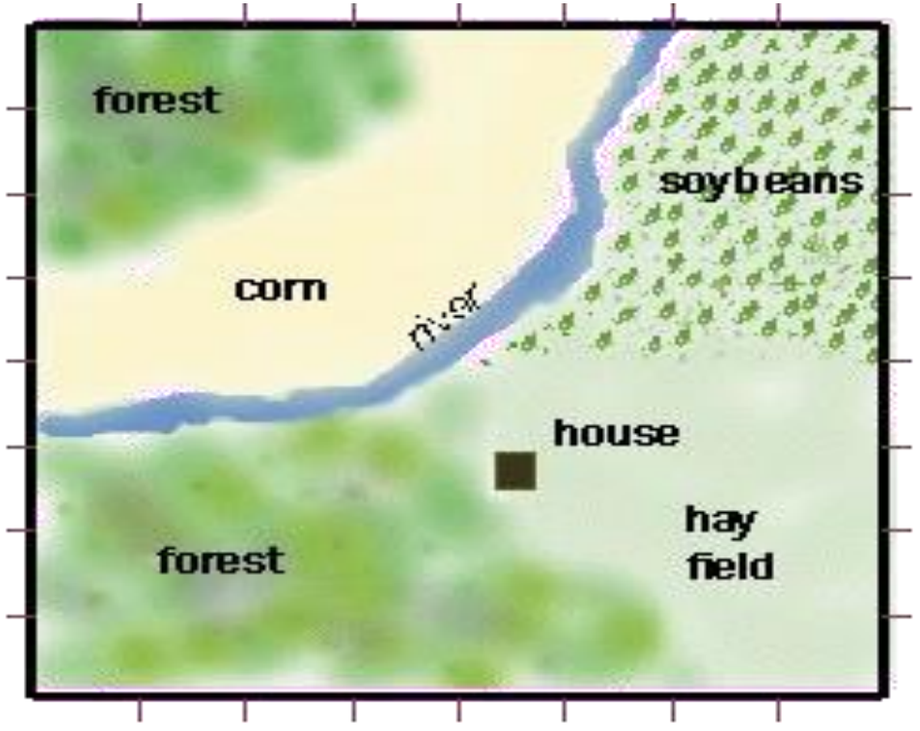


GIS – Spatial Data : Representation

RASTER	ACTUAL	VECTOR
		
		
		
		
		

GIS – Spatial Data : Representation

- Coding/Representation of Vector in GIS



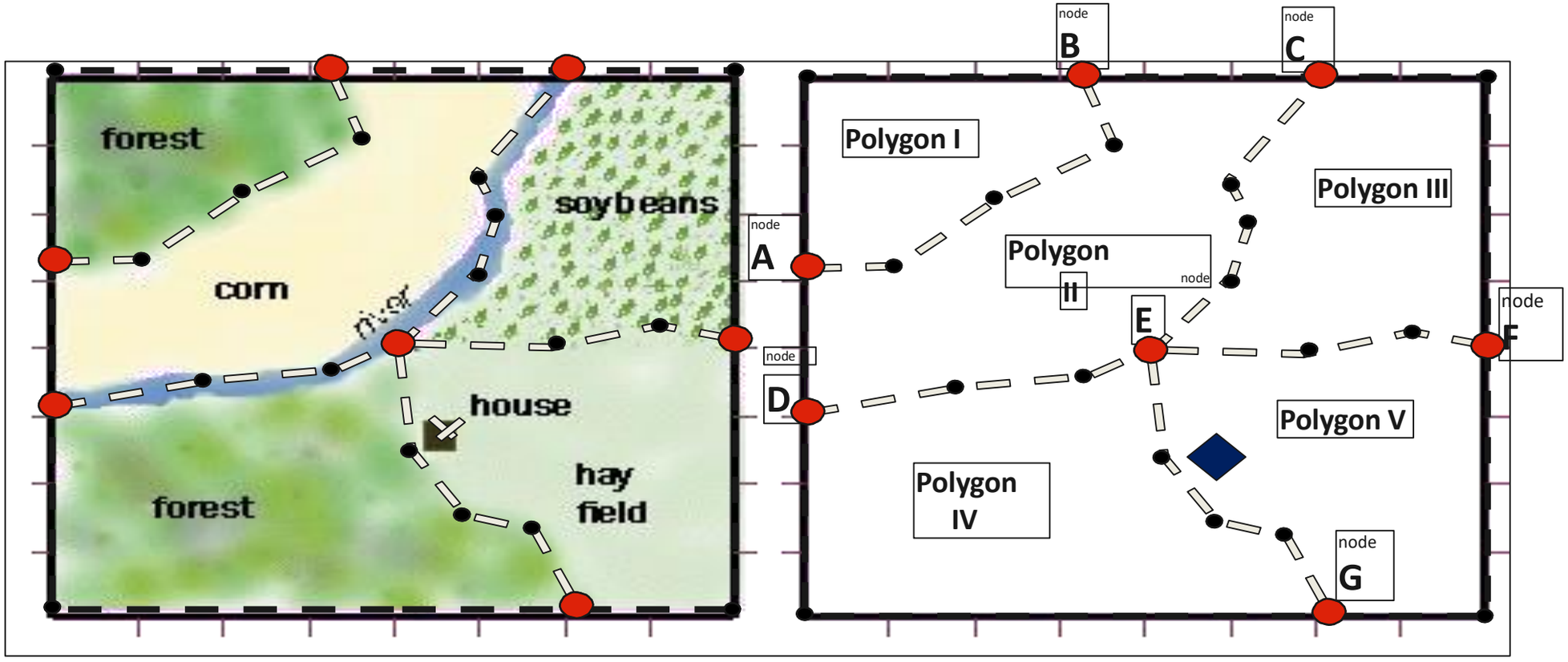
Reality



Vector Mode Model of Reality

GIS – Spatial Data : Representation

- Coding/Representation of Vector in GIS

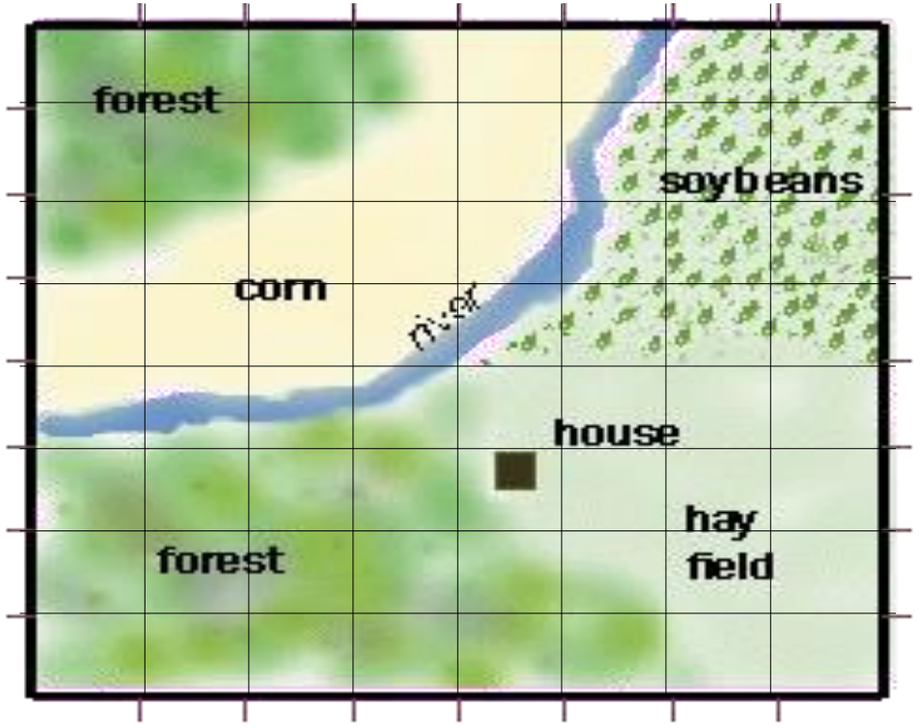


Reality

Vector Mode Model of Reality

GIS – Spatial Data : Representation

- Coding/Representation of Raster in GIS



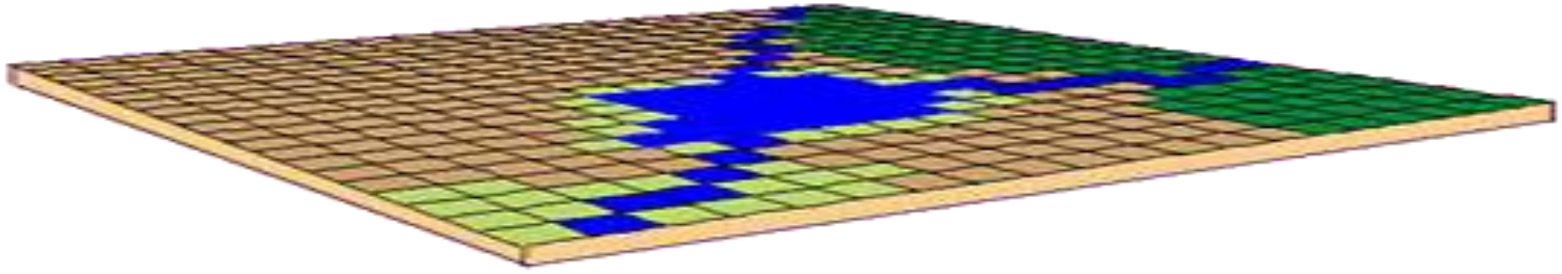
Reality

1	1	1	1	2	3	4	4
1	1	1	2	2	3	4	4
1	2	2	2	3	3	4	4
2	2	2	3	3	4	4	4
3	3	3	3	5	5	5	5
1	1	1	1	6	5	5	5
1	1	1	1	1	5	5	5
1	1	1	1	1	1	5	5

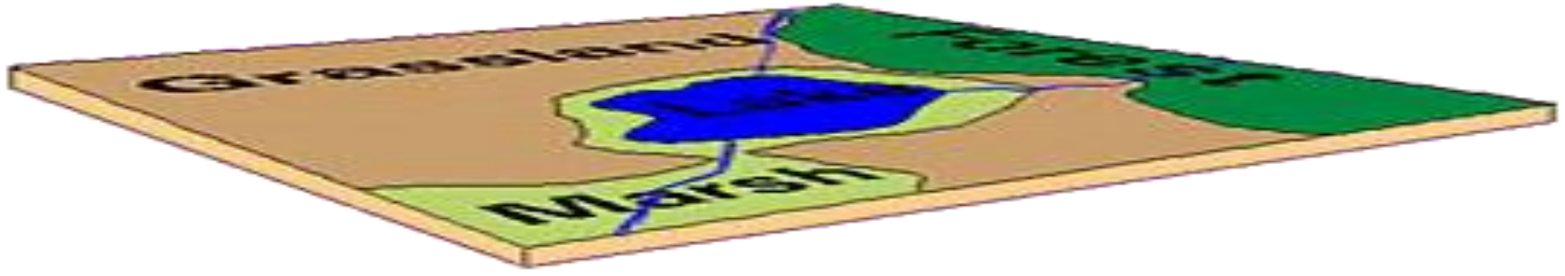
Raster Mode Model of Reality

GIS – Spatial Data : Representation

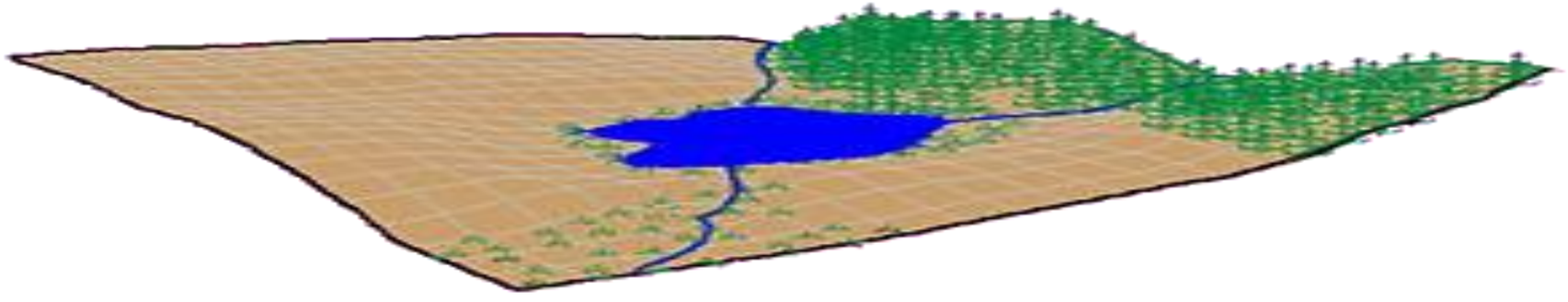
Raster Model



Vector Model

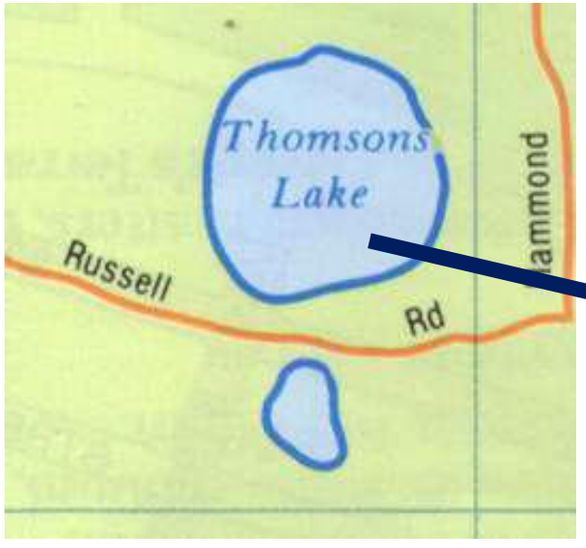


Real World



GIS – Attribute Data

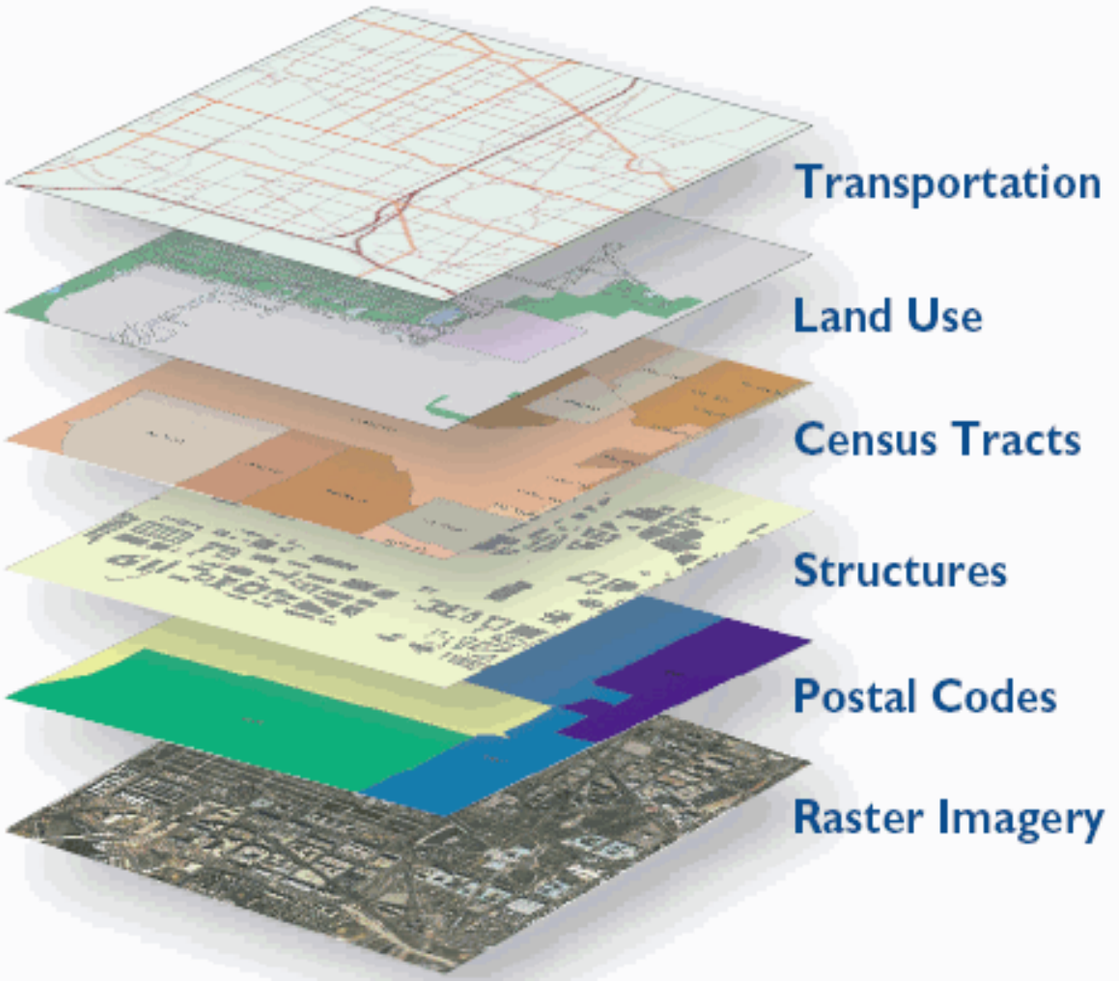
- Attributes can be **numeric or alphanumeric data** that is ***assigned to a point, line or area spatial features***
- Attribute values/data of lake may be
  - ✓ **Lake Name**
  - ✓ **Capacity**
  - ✓ **Water Quality**
  - ✓ **Depth**
  - ✓ **Area**



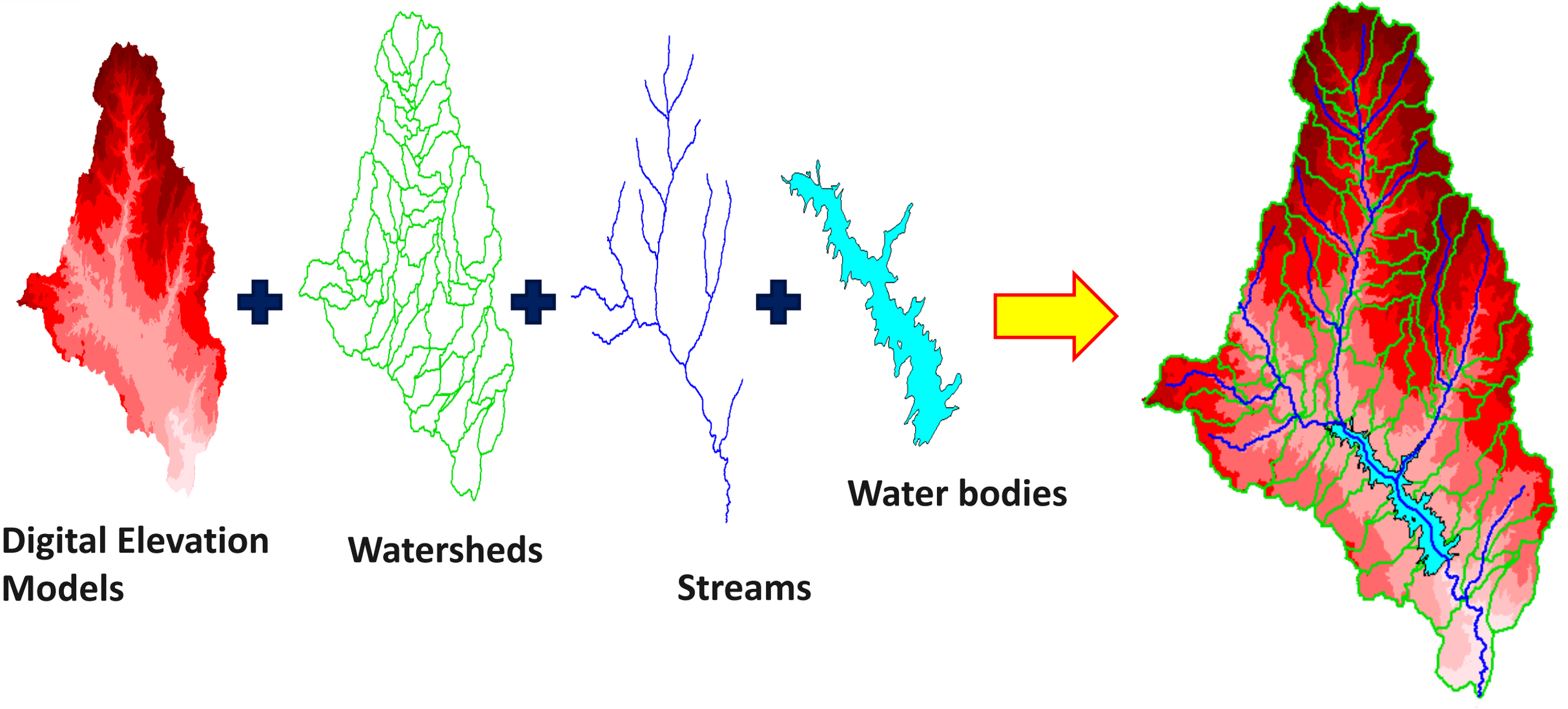


GIS – Importance of Layer

- **Geographic data = Representation of reality**
- ***Reality is complex***.
- GIS utilizes a **layer approach**
- **Each layer** only includes **information** about **one type of phenomenon**.



GIS – Importance of Layer



GIS – Answers……

- Location: What is at...? Where is it…?
  - ✓ (Location question ; what exists at a particular location)
- Condition: Status of features…?
  - ✓ (Conditional question ; which locations satisfy certain conditions)
- Trends: What has changed since...?
  - ✓ (Trendy question ; identifies geographic occurrence or trends that have changed or in the process of changing)
- Patterns: What spatial patterns exist…?
  - ✓ (Relational question : analyzes the spatial relationship between objects of geographic features)
- Modeling: What if…?
  - ✓ (Model based question ; computers and displays an optimum path, a suitable land, risky area against disasters etc. based on model)

GIS – Answers.....

Location: What is at...?

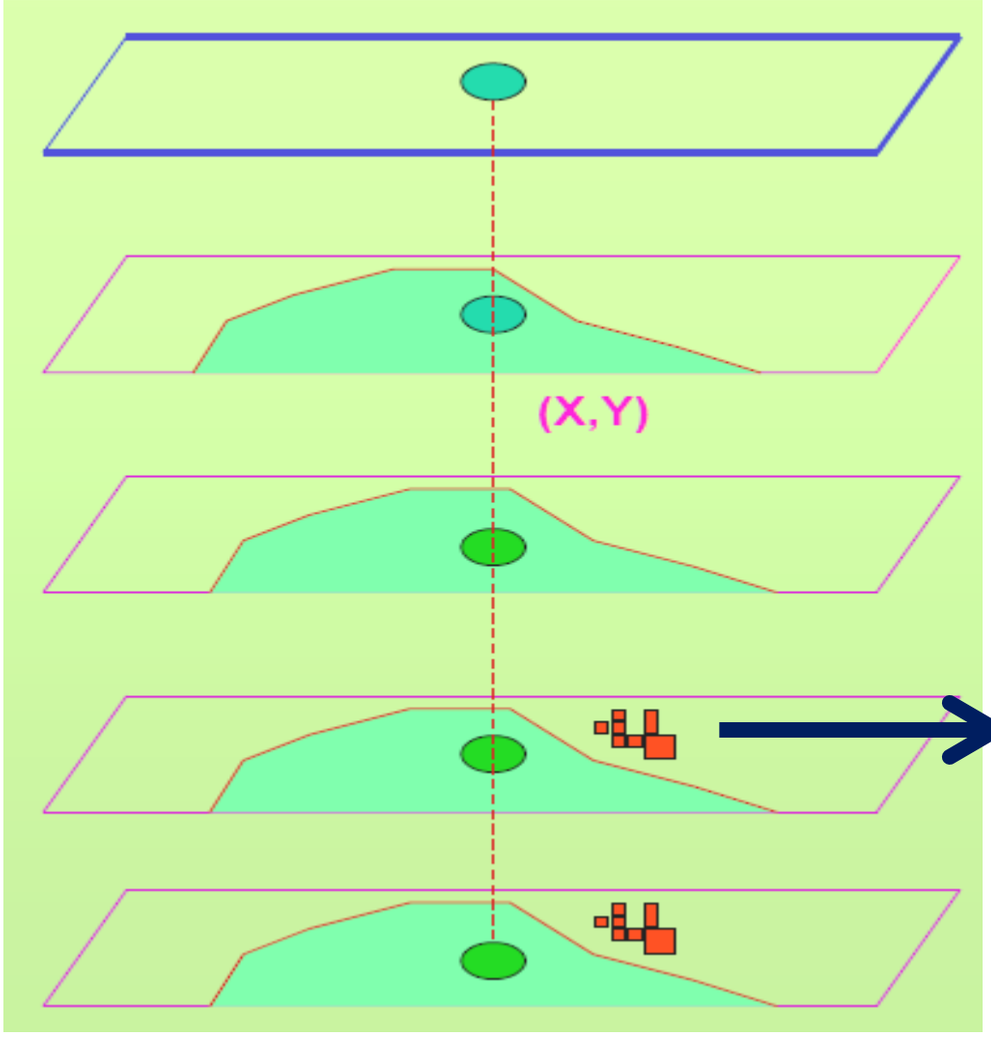
Police Colony

Where is it...?

Within City

Condition: Status of features...?

Residential



Trends: What has changed since...?

Price of land increased

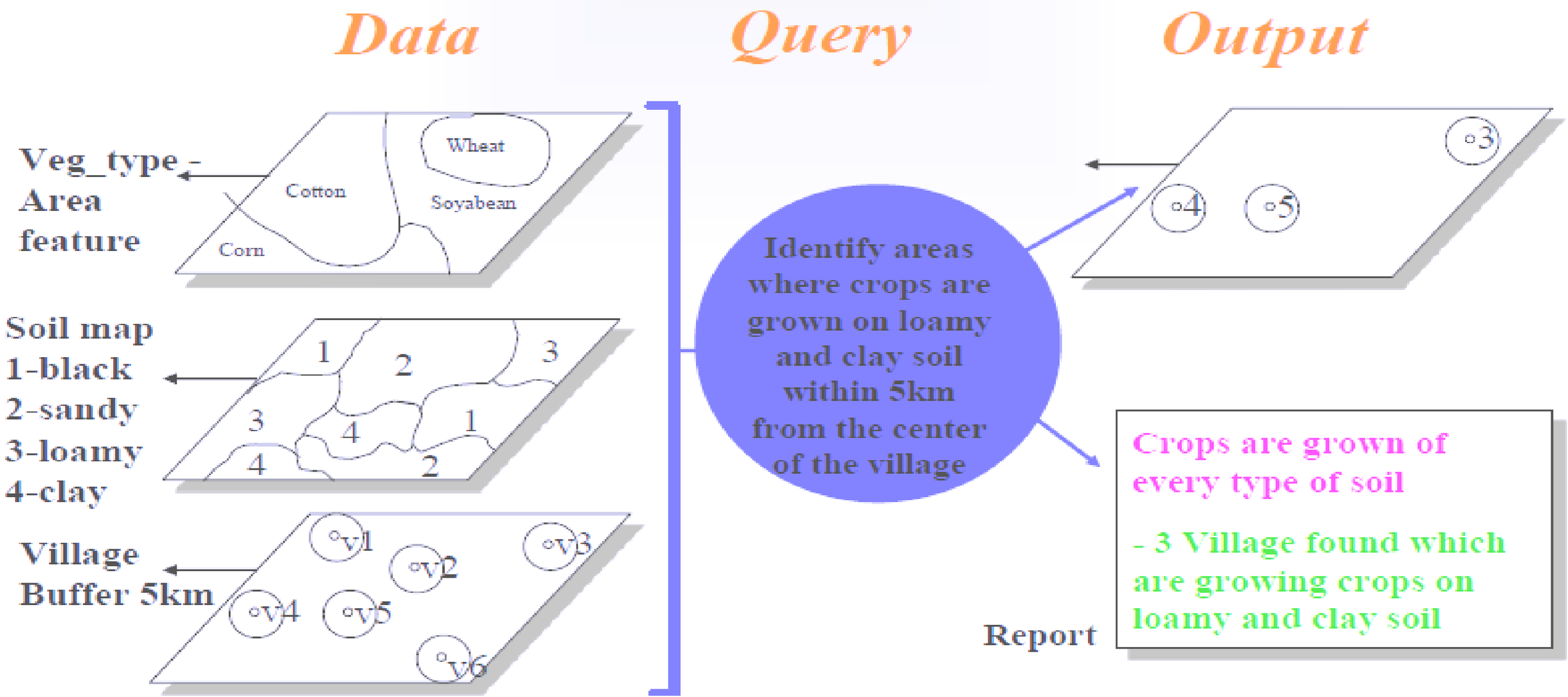
Patterns: What spatial patterns exist...? Related data ...?

Nearest Police station

Modeling: What if...?

Low lying area and hence get flooded in heavy rainfall

GIS – Applications

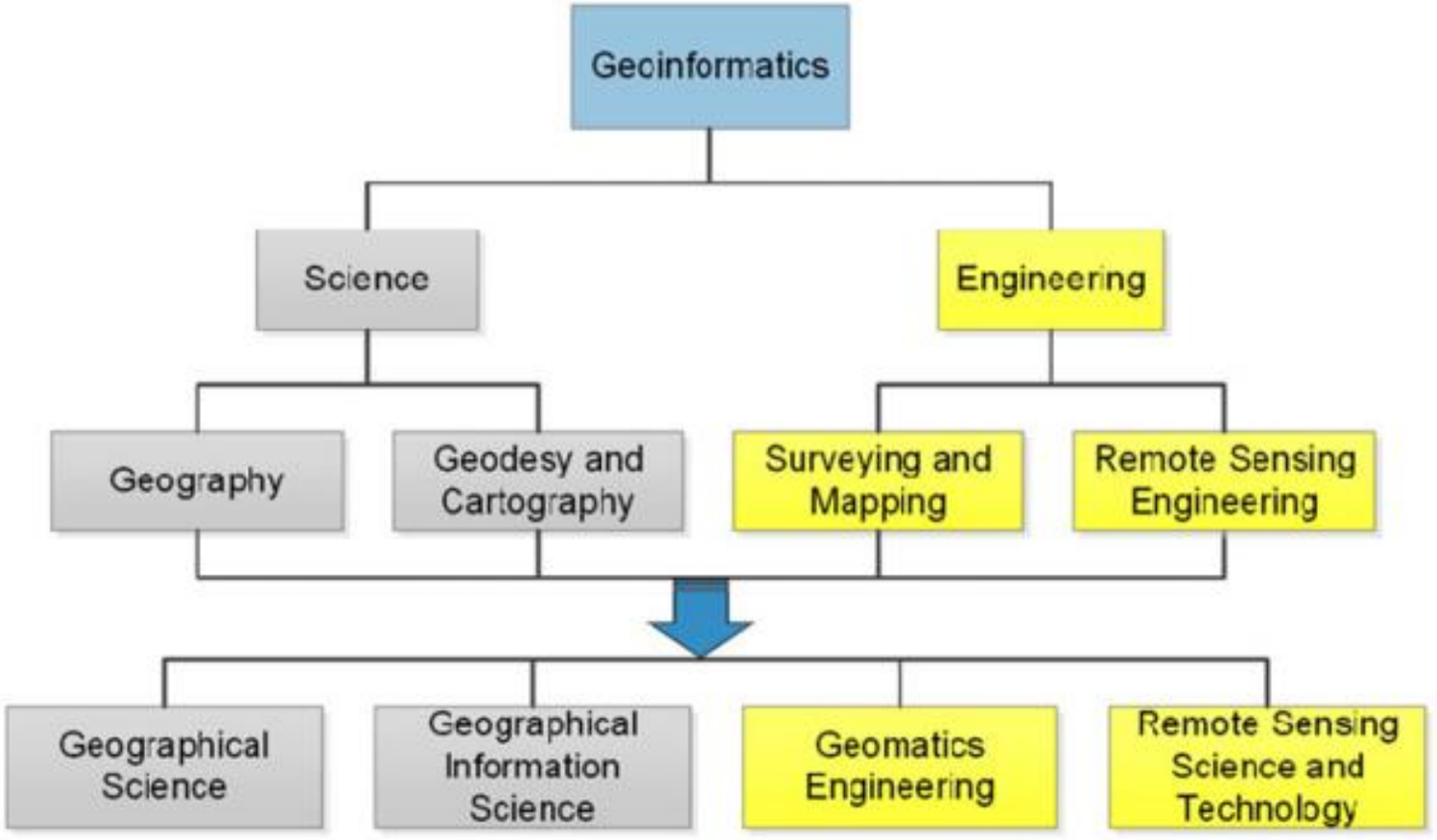


What Is Geoinformatics?

- **Geoinformatics** is the science and the technology which develops and uses information science, infrastructure to address the problems of geography, geosciences and related branches of engineering.

- *"The art, science or technology dealing with acquisition, storage, processing, production, presentation and dissemination of geoinformation"*

What Is Geoinformatics?



GIS – Why????

We Live in Two Environments

Natural



Self-Regulated

Manmade



Managed

And they are increasingly in conflict

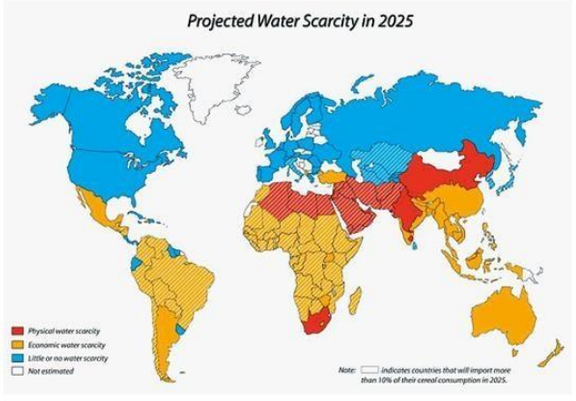
GIS – Why????

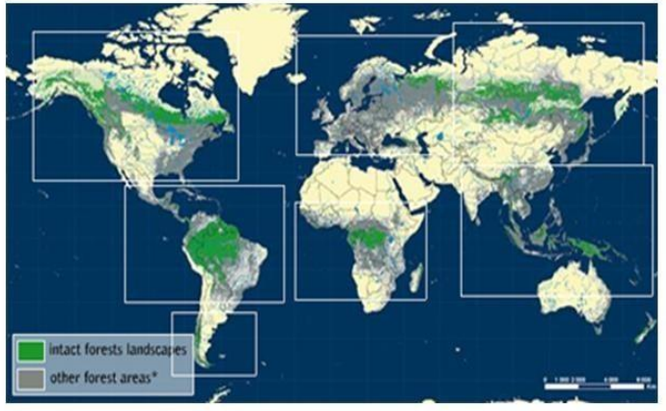

We Use Many Languages to Describe Our Environment
They Reflect Our Experiences
We use abstractions...
Cartography
Mathematics
Models
Photography
English
汉语
한국어
Ελληνικά
Français
Written
Taxonomy
Music
Art
Diagrams
Statistics
Software
...To Record and Share
...And Organize Our Reality

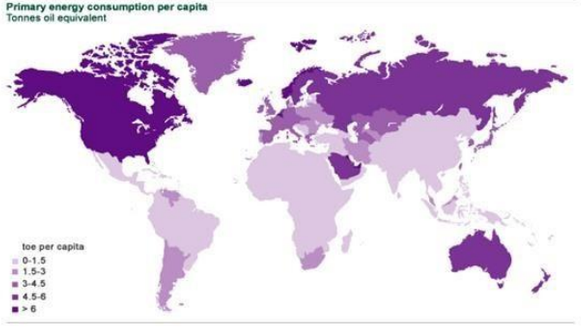
GIS – Why????

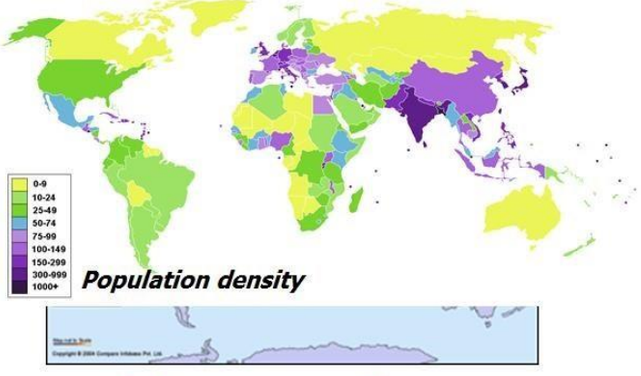
Our Environment is Changing Rapidly

Becoming Increasing Challenged









...Influenced by everything we do...

...and also don't do

GIS – Why????

Driven by the Dynamics of

Increasing

Population

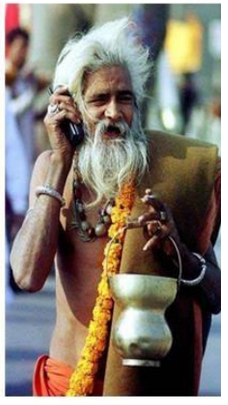
Resource Consumption

Development and Globalization

Political and Social Controversy

Technological Change












Threatening

Our Climate

Our Biodiversity

Our Natural Places

Our Stability and Security

Our Health

…affecting the possibility of our future

…sustainability of our environment

GIS – Why????

And we are Evolving towards a more human controlled environment

Burgeoning

Populace

Energy exhaustion

Industrialization

Pollution

Dying

Resources

Natural Areas

Diversity

Civilization



..And creating problems in both the Environments

GIS – Why????

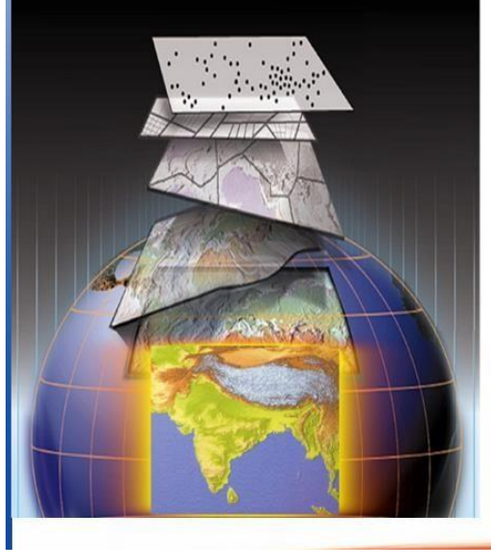
To meet these challenges

We Need to have a common Vision: Building a Digital Abstraction of the Earth

We Need Better Ways To...

- **Represent**
- **Understand**
- **Manage**
- **Communicate**

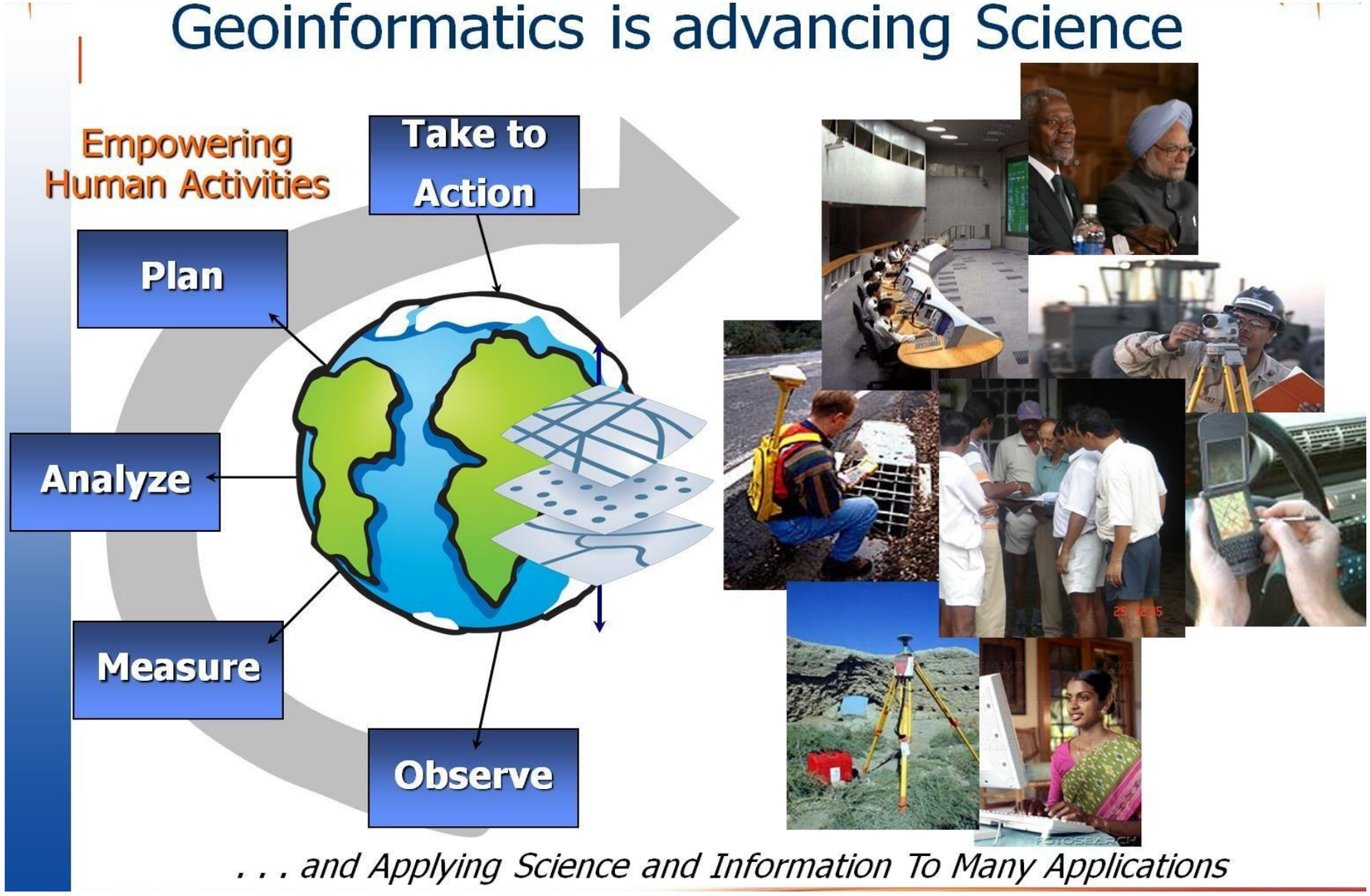
....Our World As a System





...Geoinformatics provides the framework

GIS – Why????

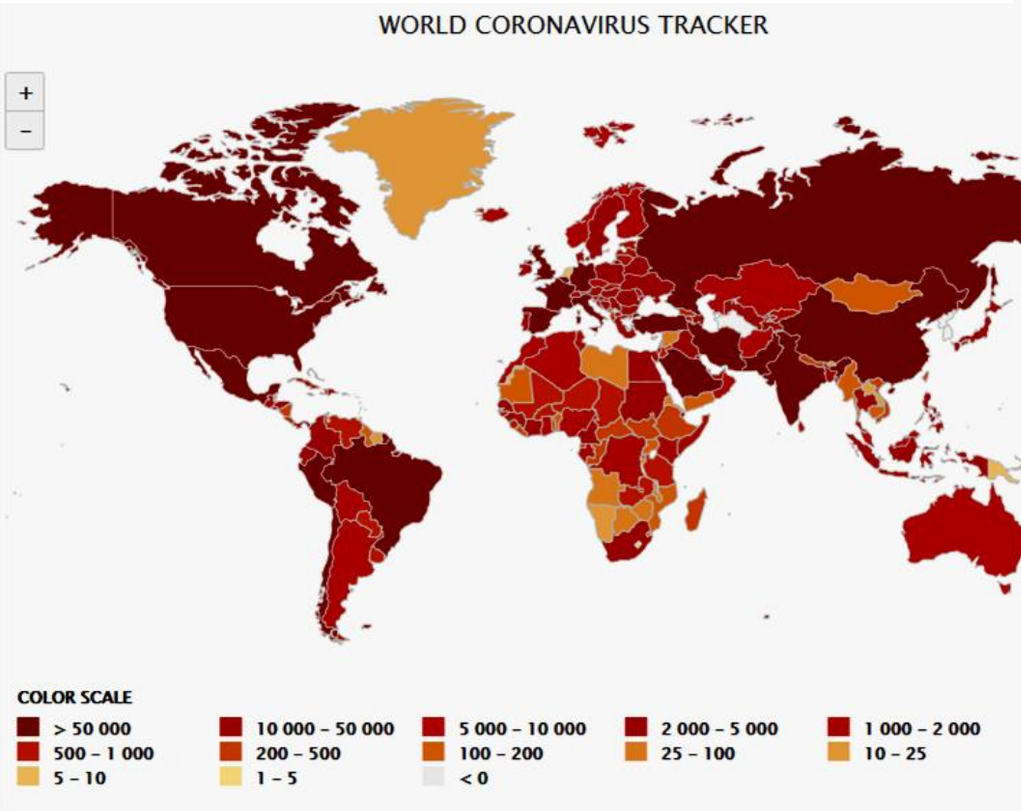

Geoinformatics is advancing Science
Empowering Human Activities
Take to Action
Plan
Analyze
Measure
Observe
. . . and Applying Science and Information To Many Applications

GIS – Why????

Examples of Applied GIS

- **Urban Planning, Management & Policy**
  - Zoning, subdivision planning
  - Land acquisition
  - Economic development
  - Code enforcement
  - Housing renovation programs
  - Emergency response
  - Crime analysis
  - Tax assessment
- **Environmental Sciences**
  - Monitoring environmental risk
  - Modeling stormwater runoff
  - Management of watersheds, floodplains, wetlands, forests, aquifers
  - Environmental Impact Analysis
  - Hazardous or toxic facility siting
  - Groundwater modeling and contamination tracking
- **Civil Engineering/Utility**
  - Locating underground facilities
  - Designing alignment for freeways, transit
  - Coordination of infrastructure maintenance
- **Business**
  - Demographic Analysis
  - Market Penetration/ Share Analysis
  - Site Selection
- **Education Administration**
  - Attendance Area Maintenance
  - Enrollment Projections
  - School Bus Routing
- **Real Estate**
  - Neighborhood land prices
  - Traffic Impact Analysis
  - Determination of Highest and Best Use
- **Health Care**
  - Epidemiology
  - Needs Analysis
  - Service Inventory

GIS – Applications



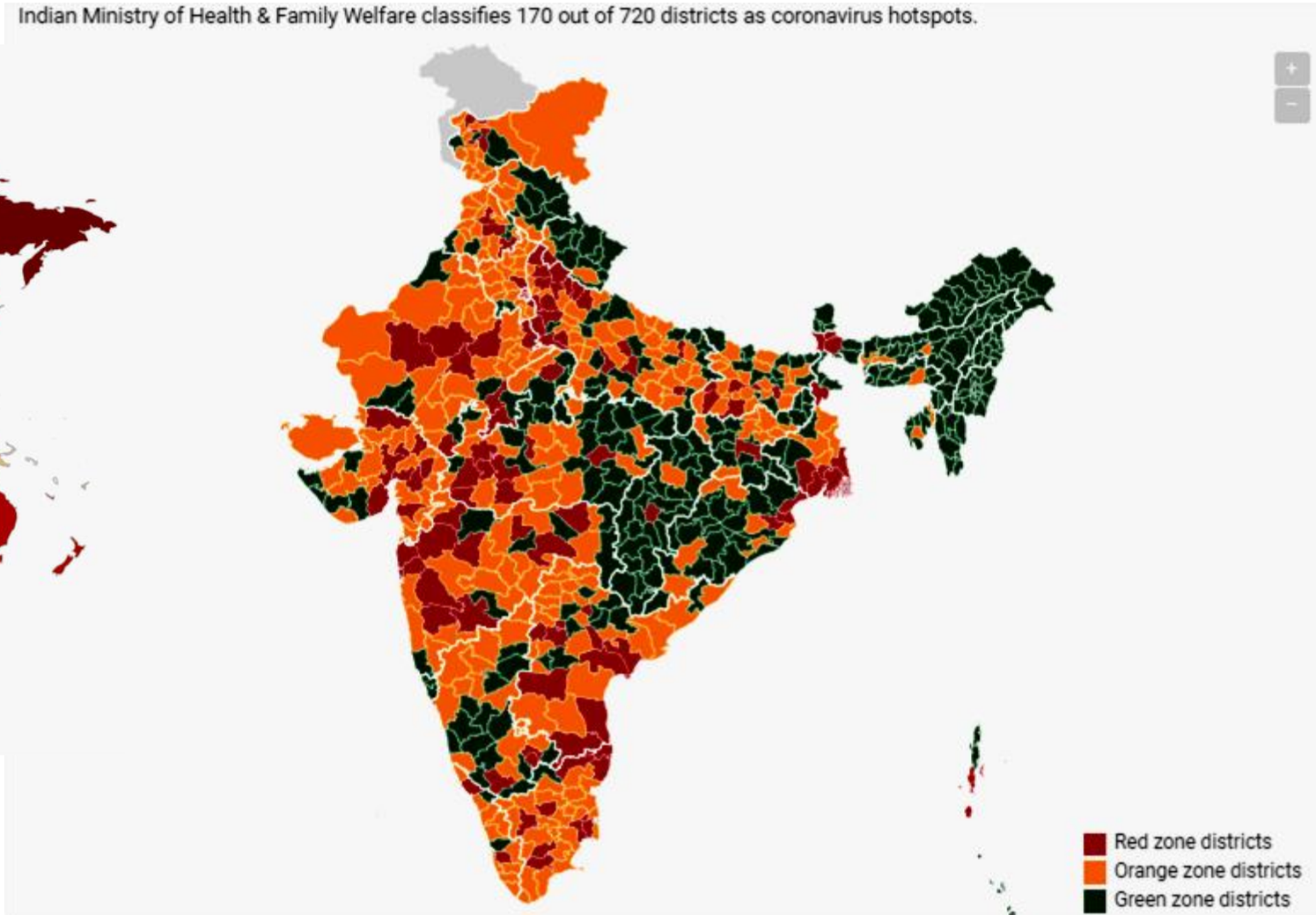


Image Source: orfnline.org, ha-asia.com

GIS – Applications

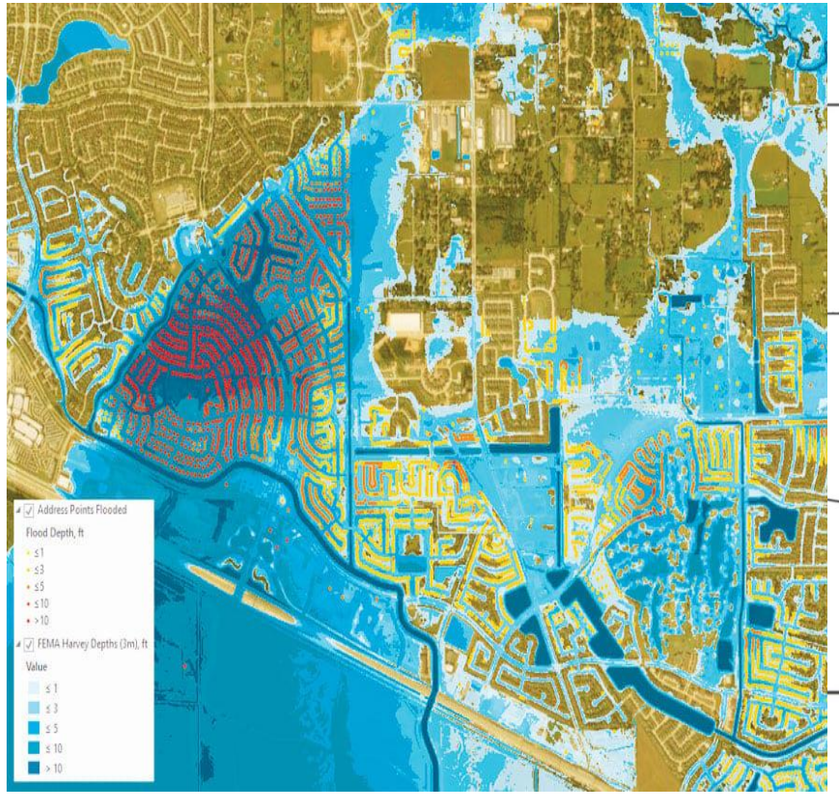
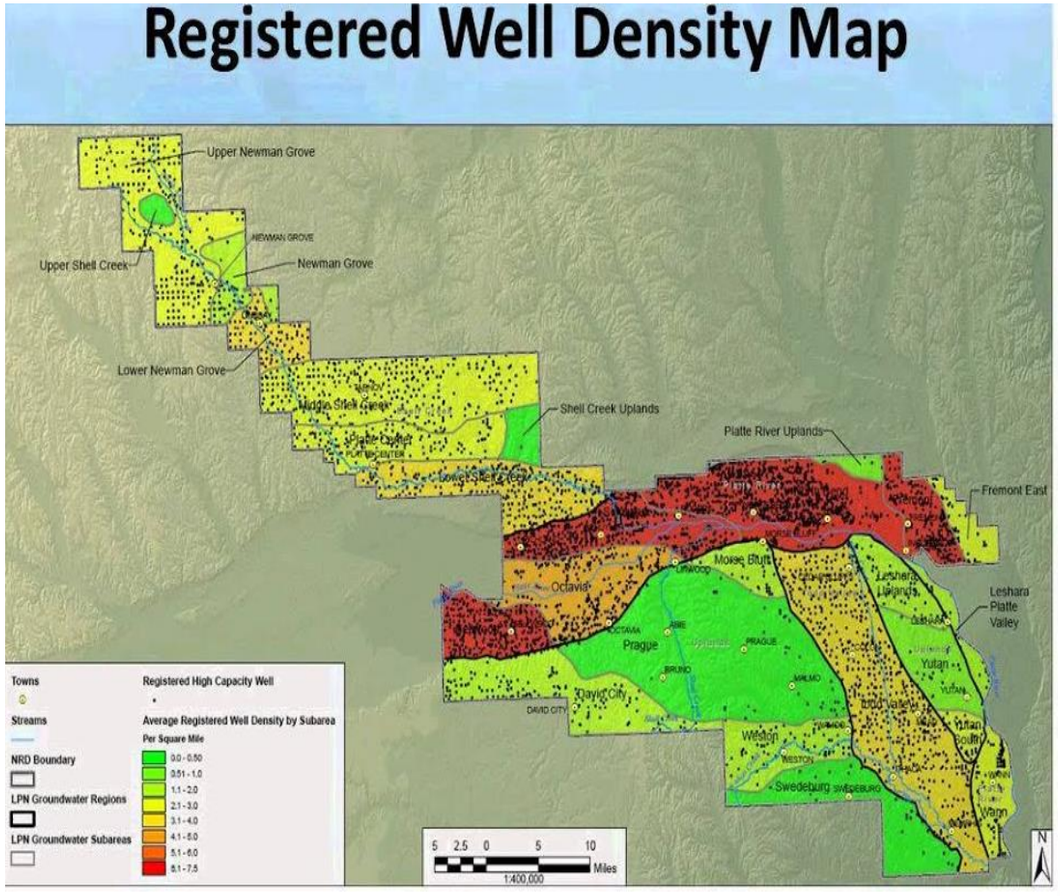




Image Source: ESRI, lahabracity.com

GIS – Applications



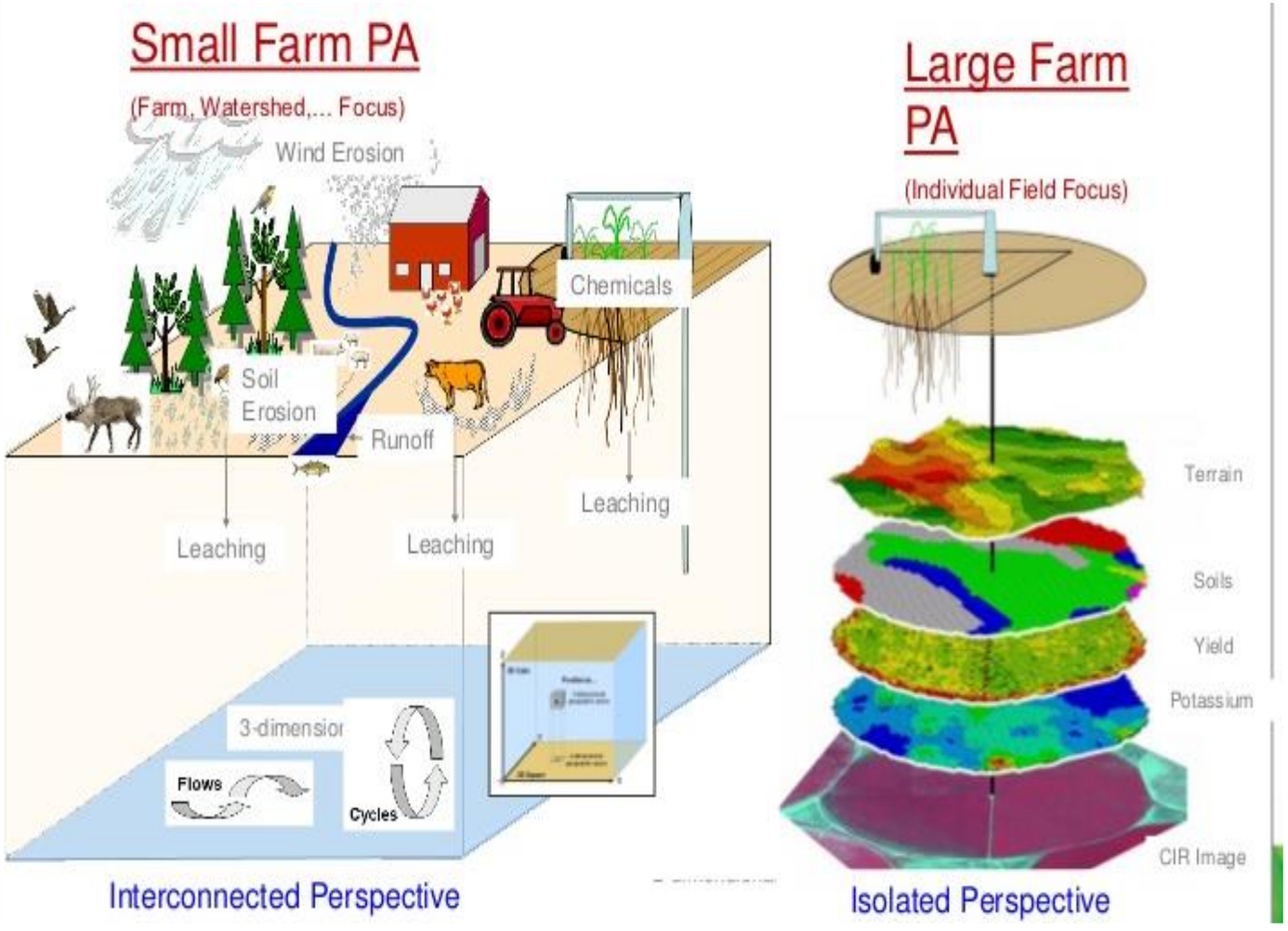
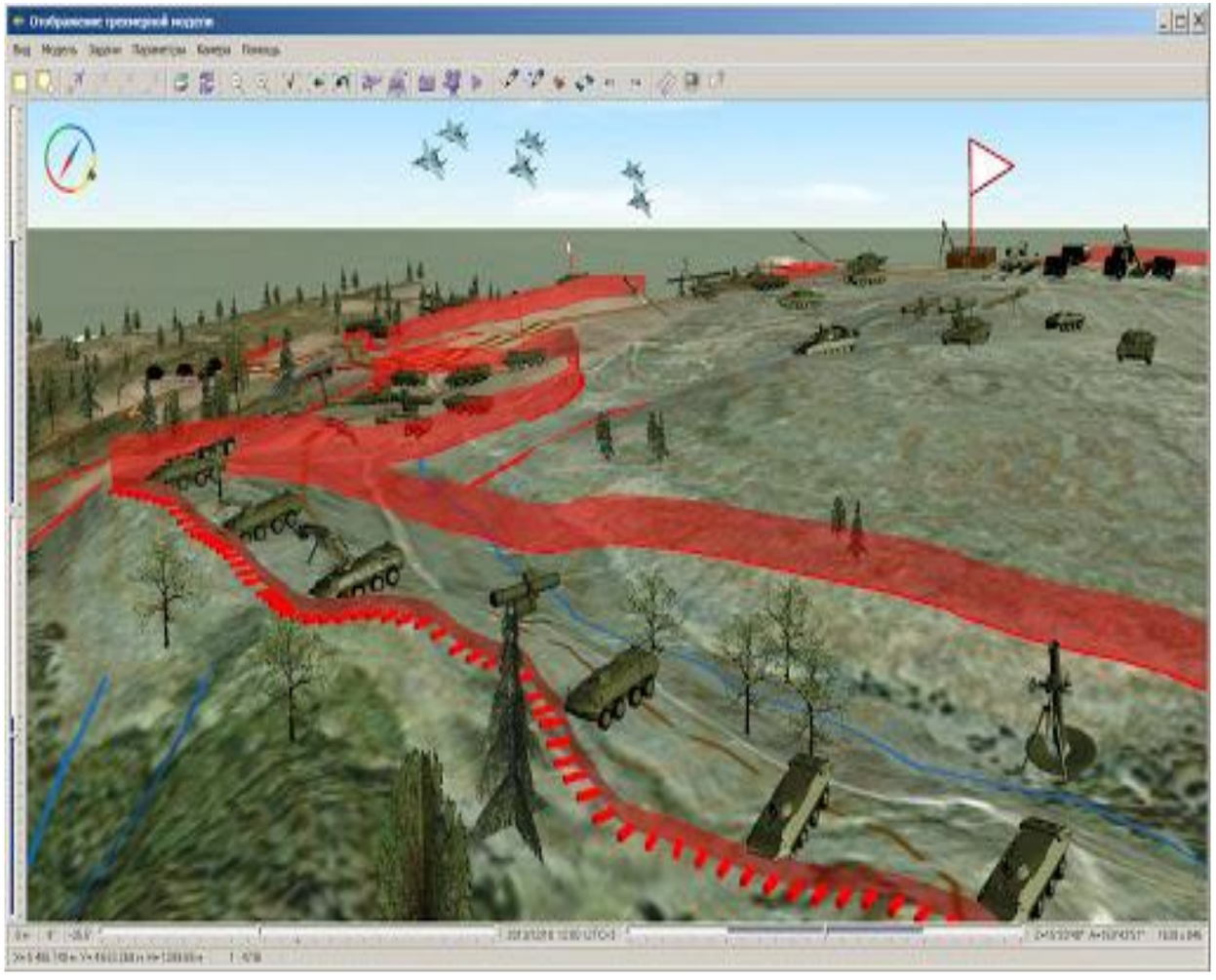


Image Source: baystreetex.com/

GIS – Applications



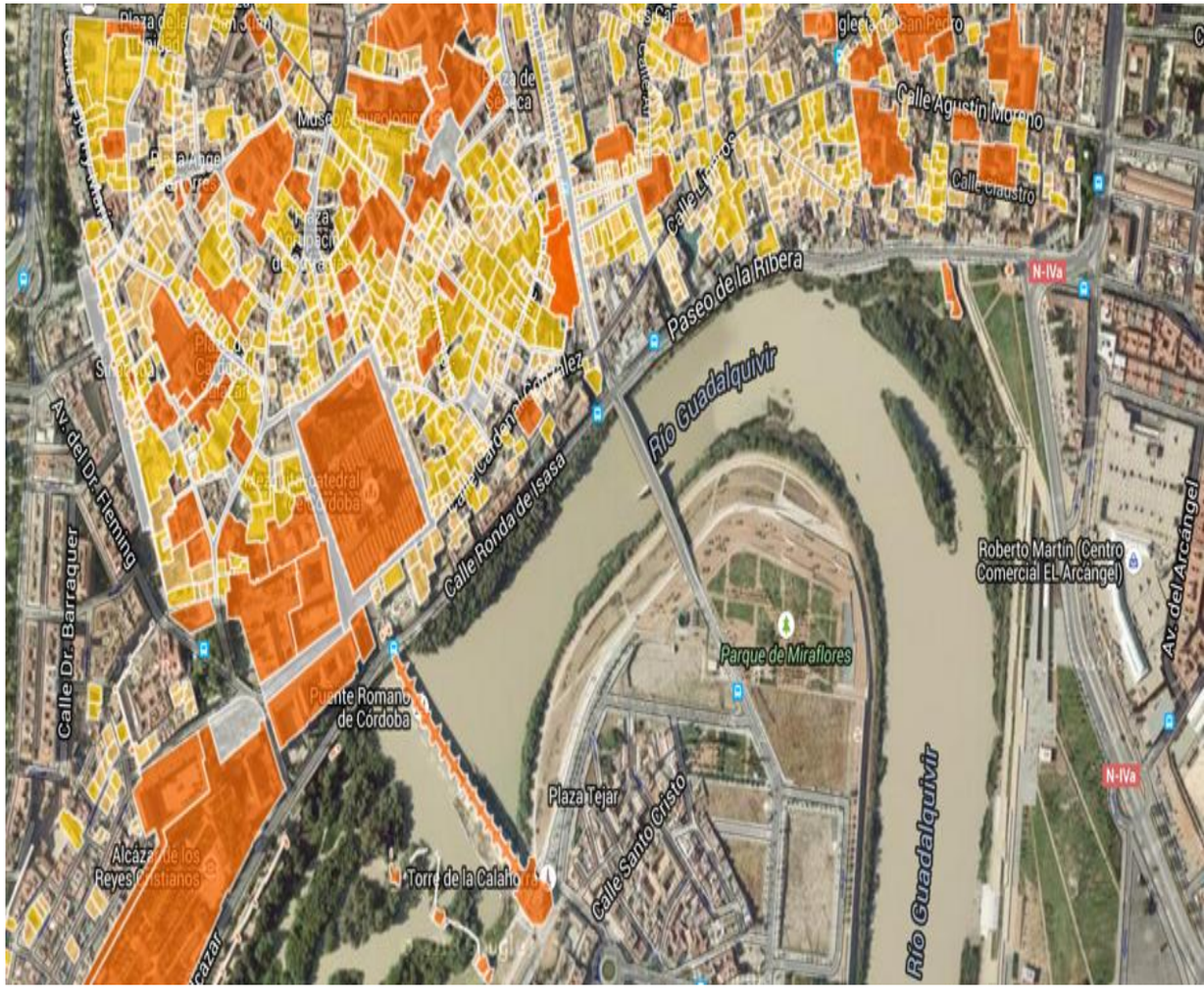
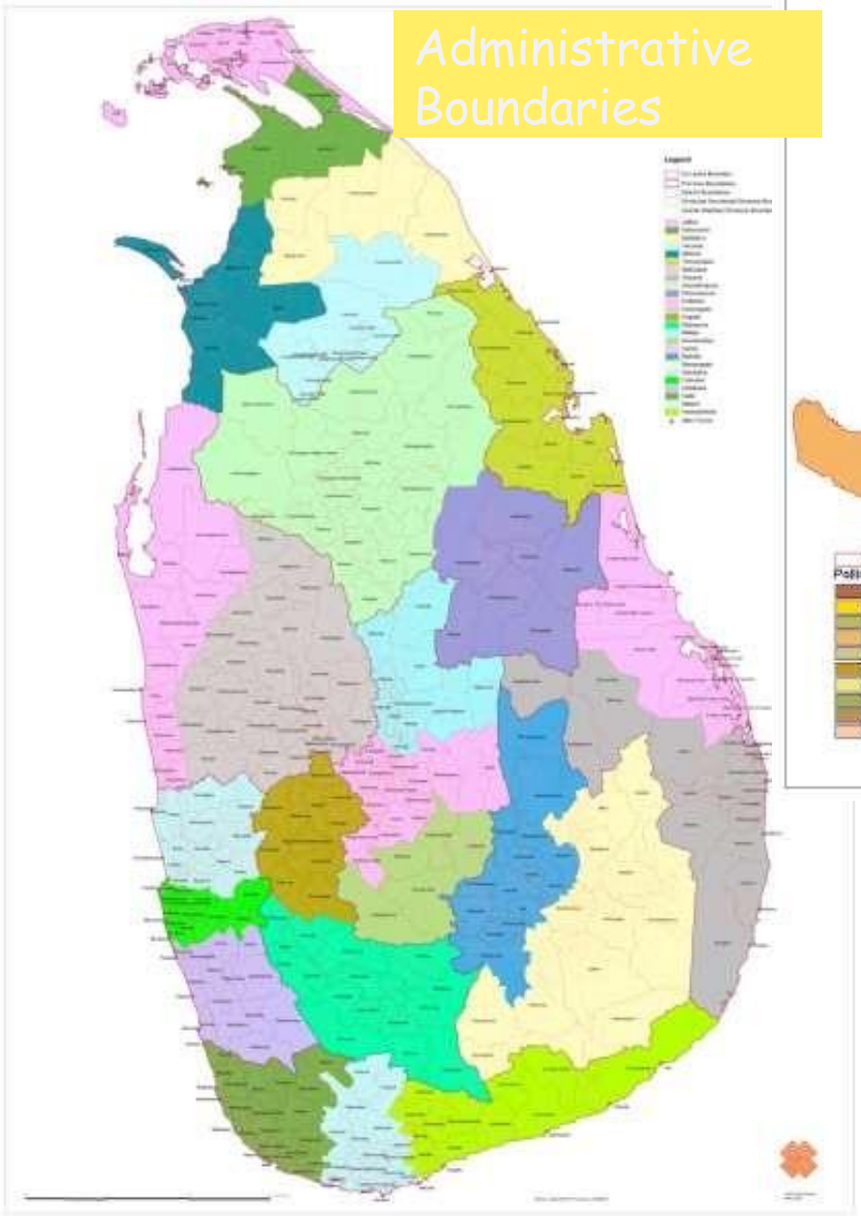
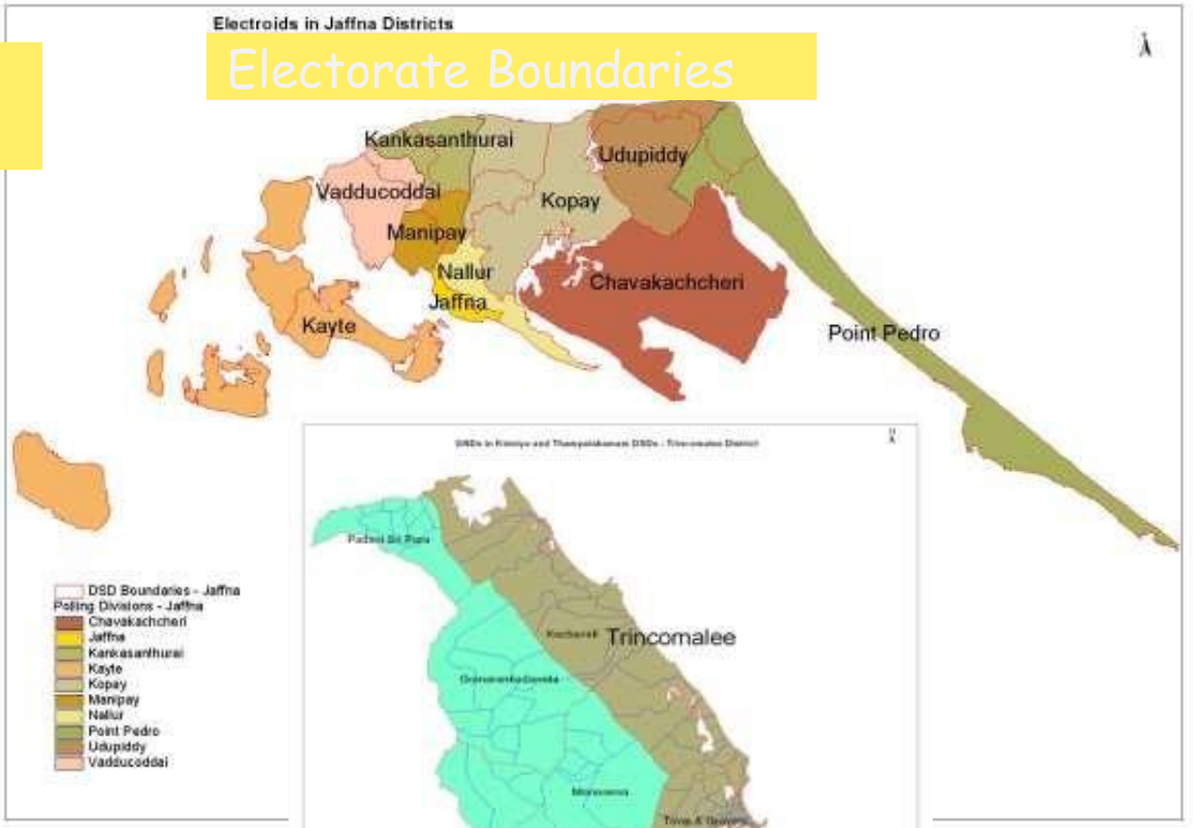


Image Source: gisinfo.net, geospatial world

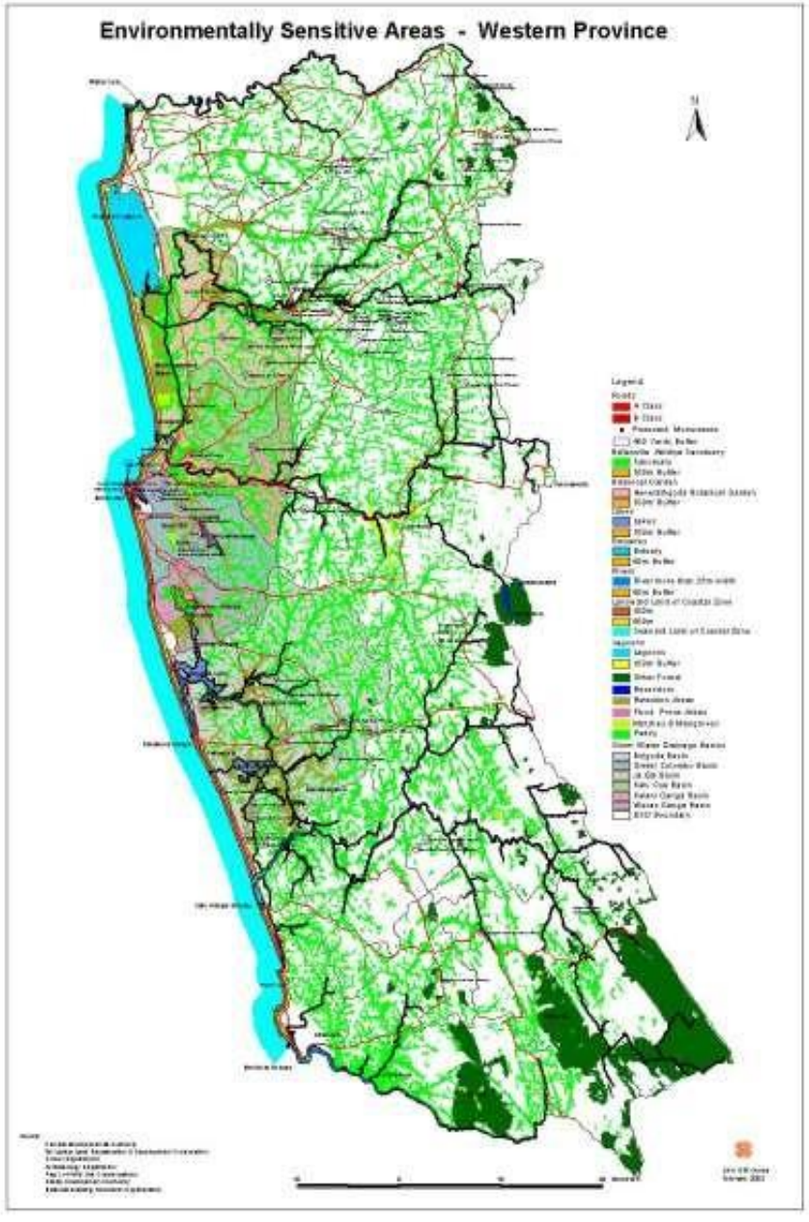
Mapping of Administrative Boundaries

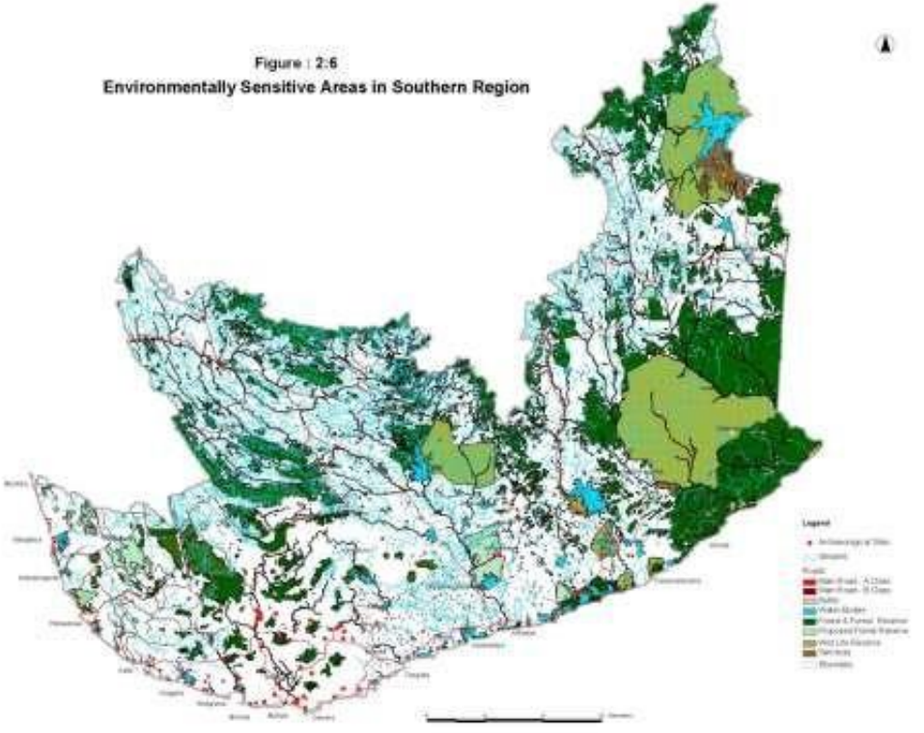






Delineating of Environmental Sensitive Areas

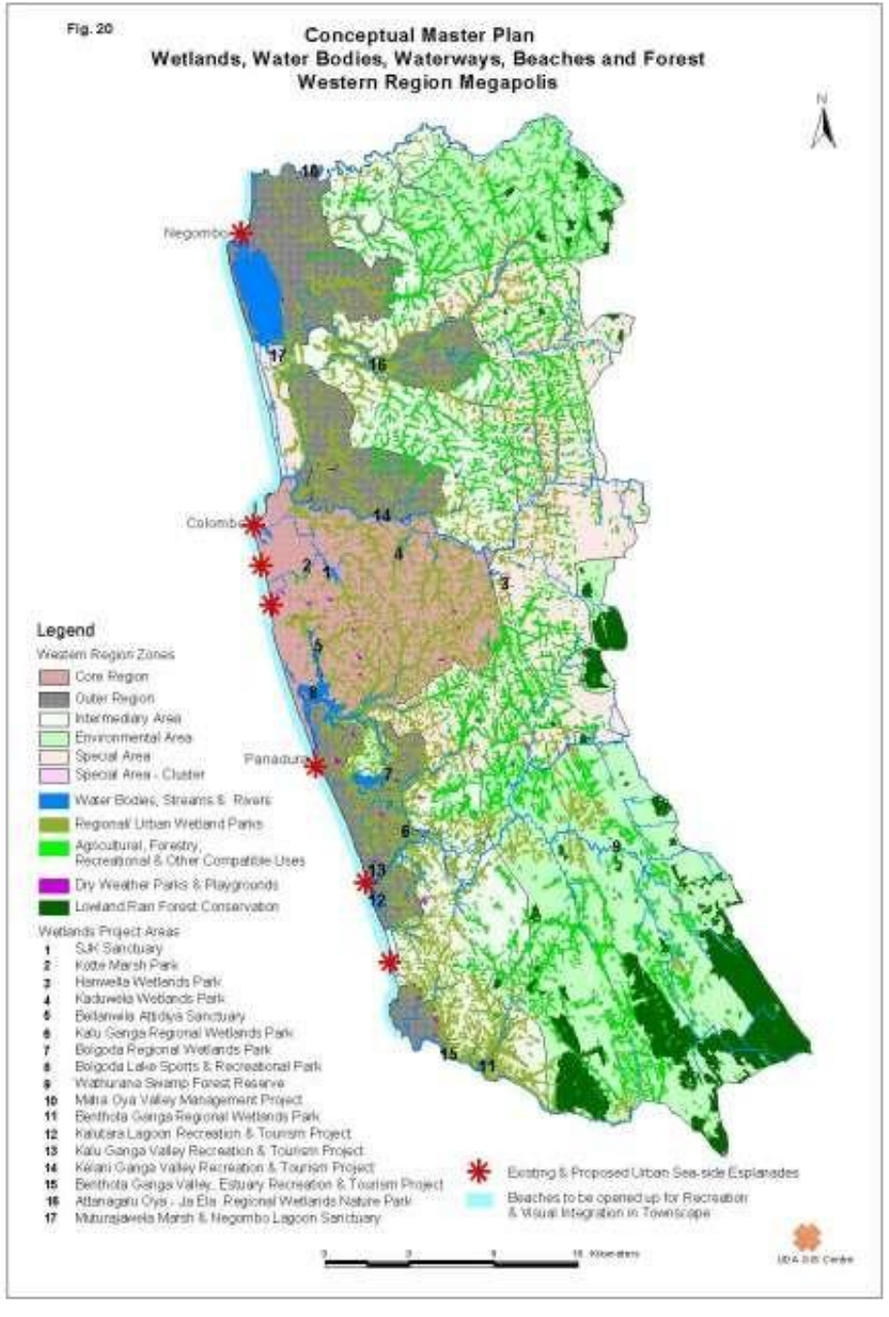


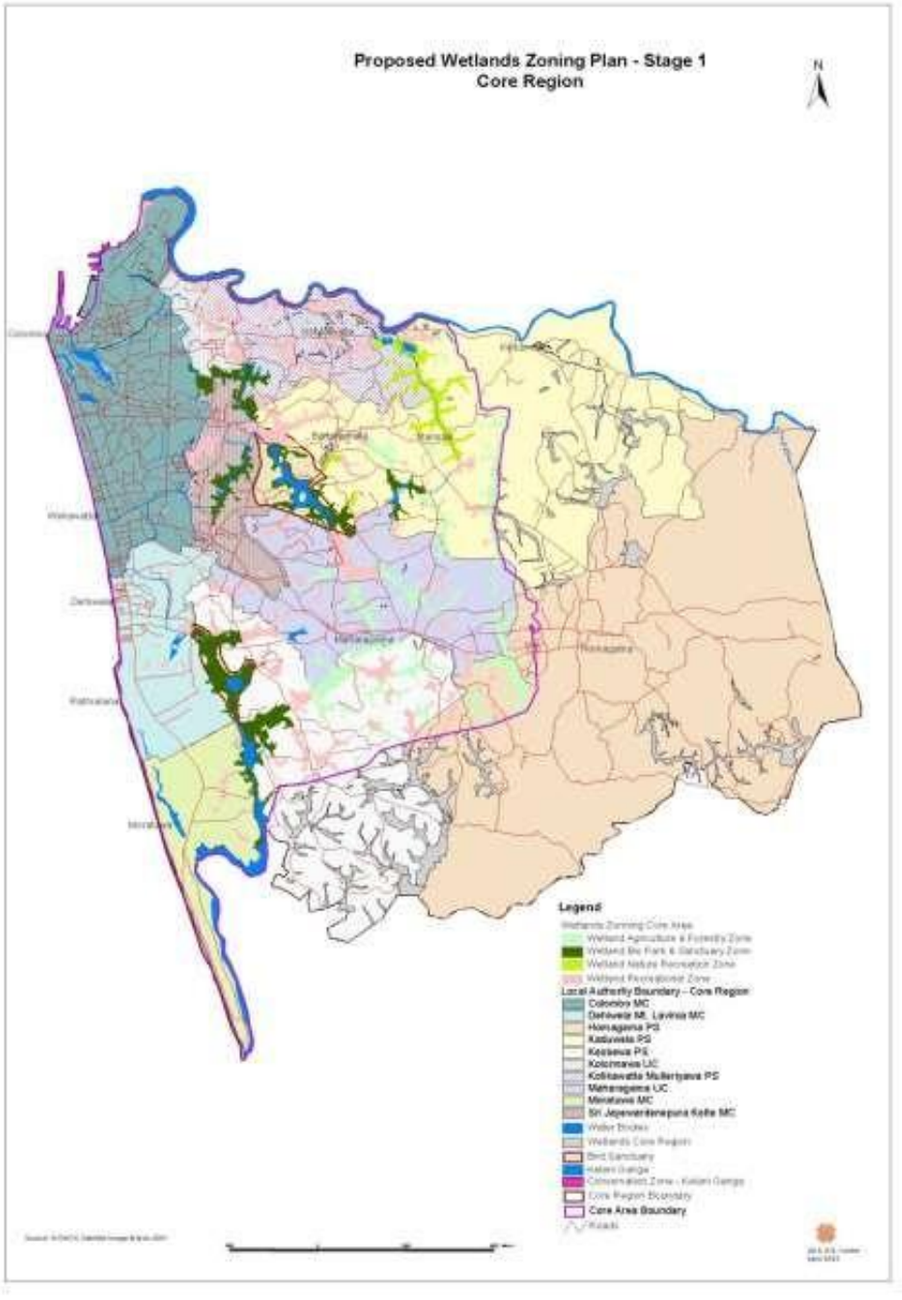


Criteria Used

1. All Water Bodies, for Rivers 60 buffer
2. All paddy and inland wetlands
3. All Roads
4. Archaeological Places, 400 metre buffer
5. All forests, Sanctuaries, national Parks

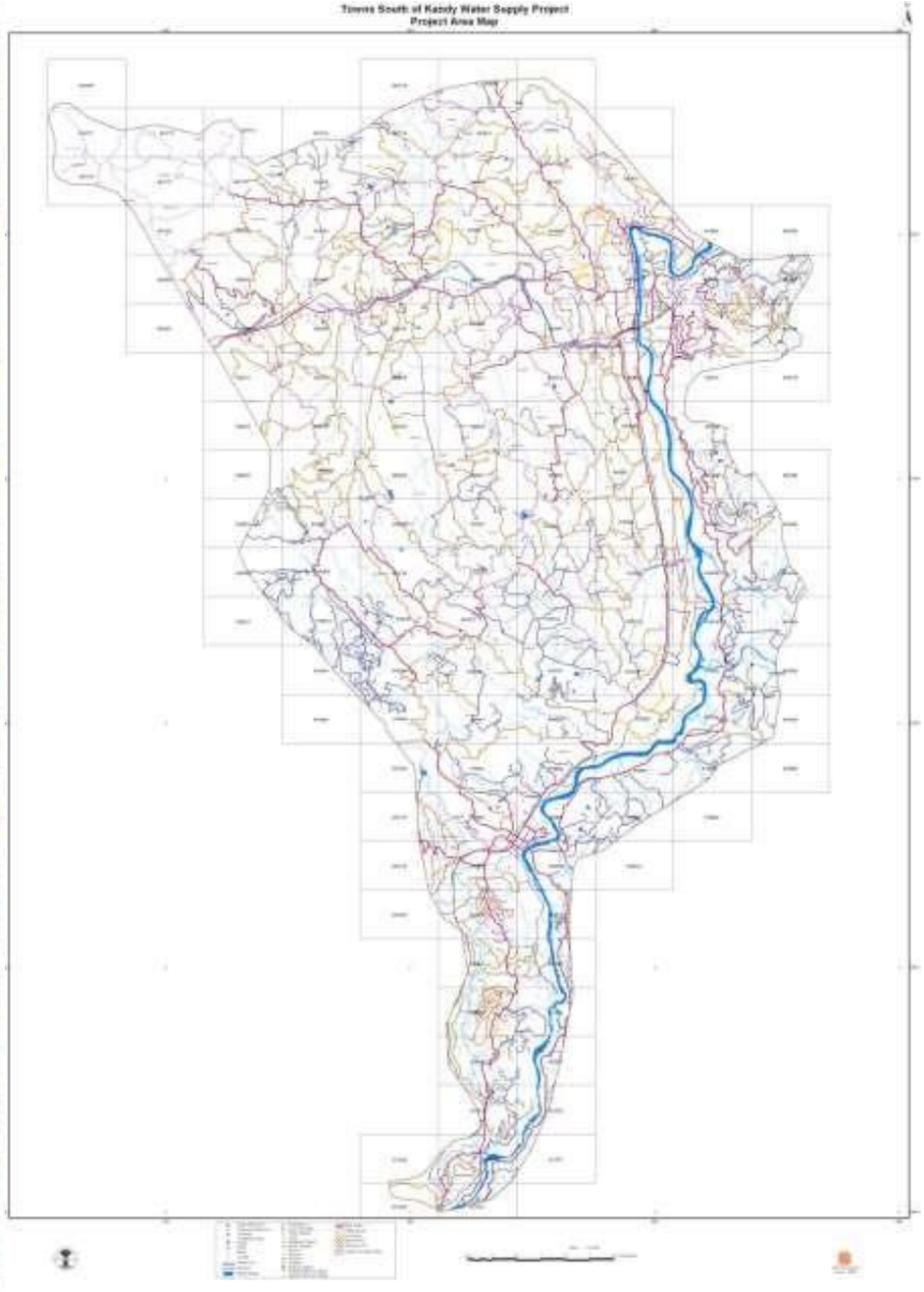
Wetlands Zoning Plans



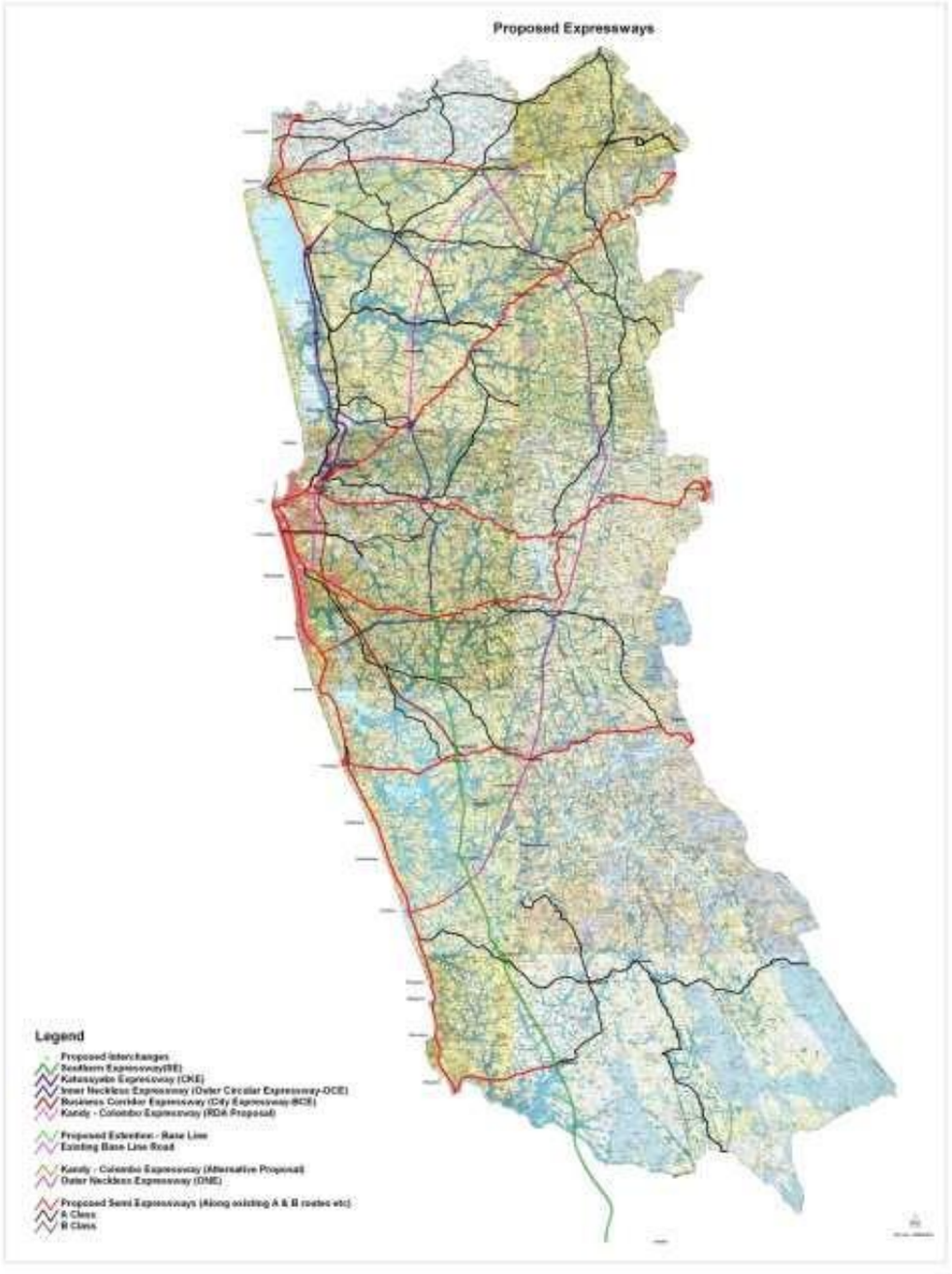


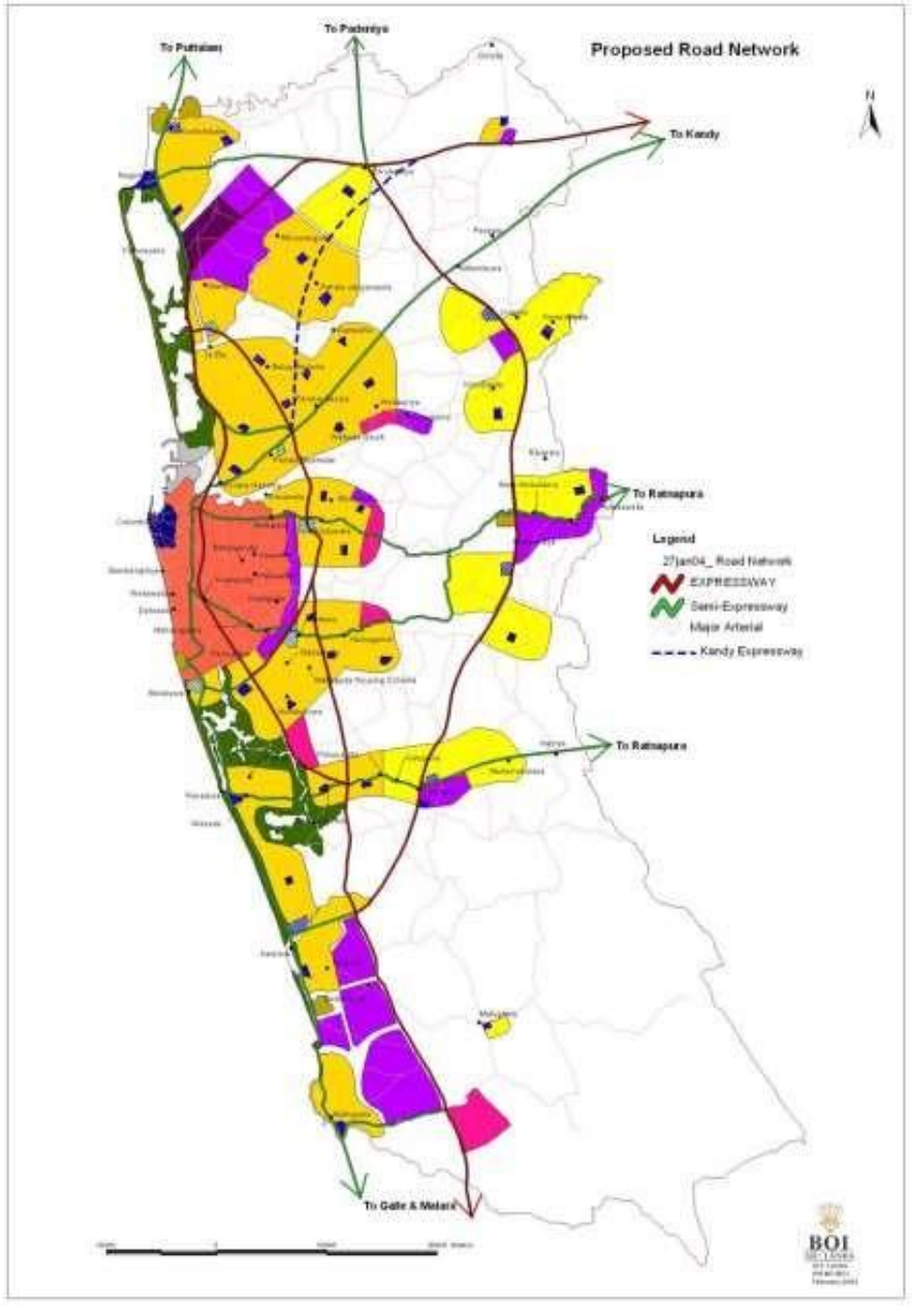
Water Demand & Supply analysis

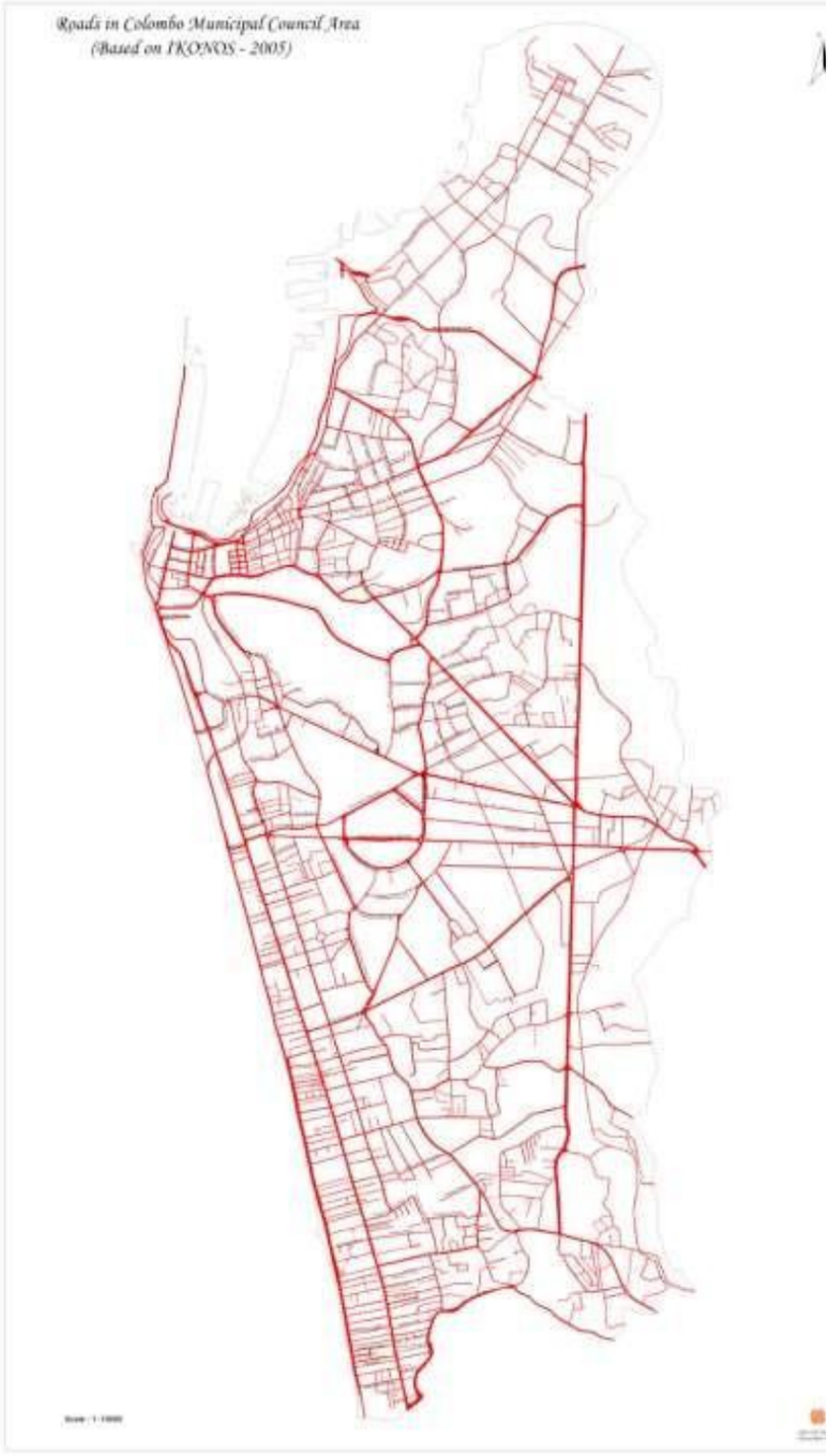


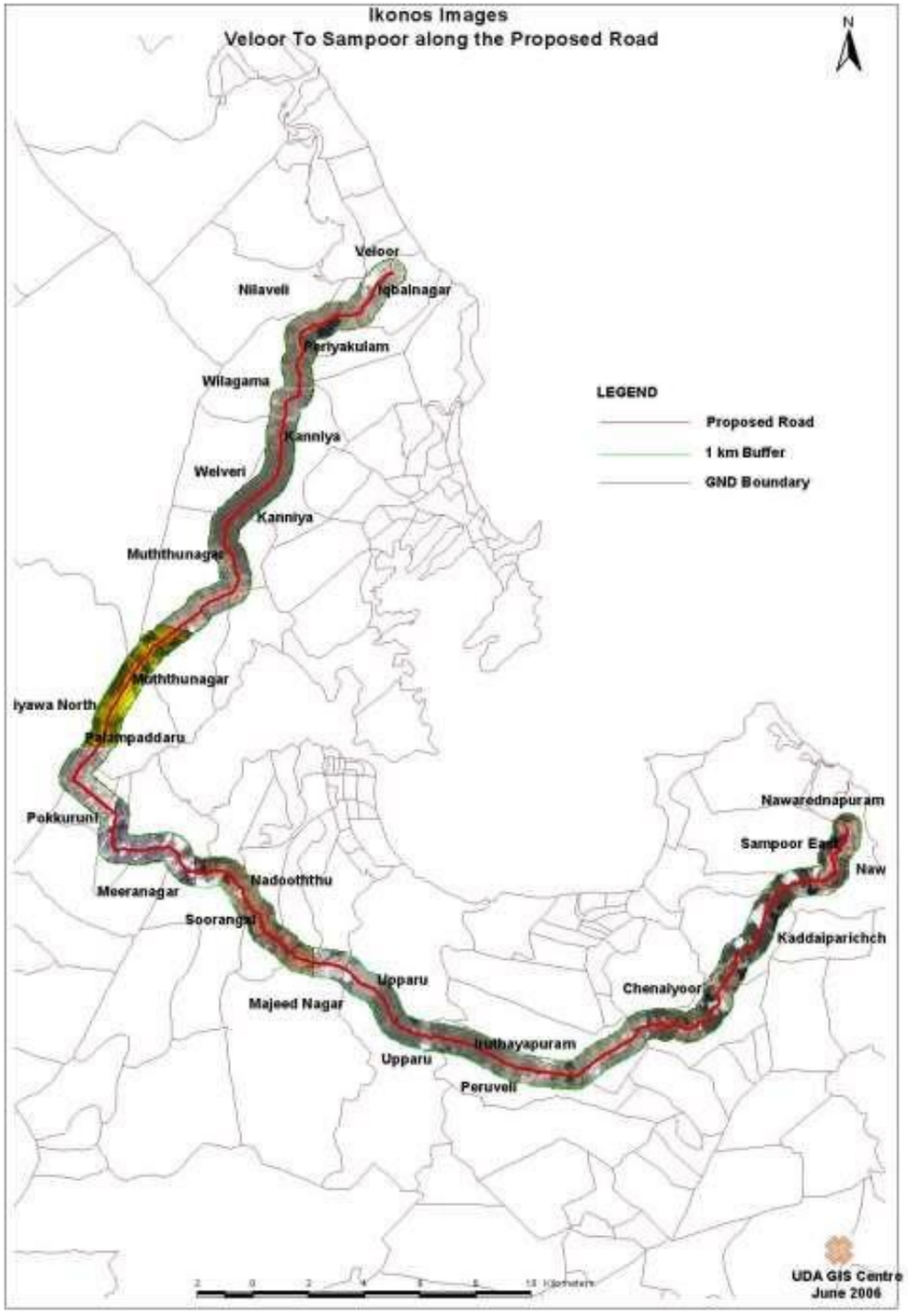


Road Network

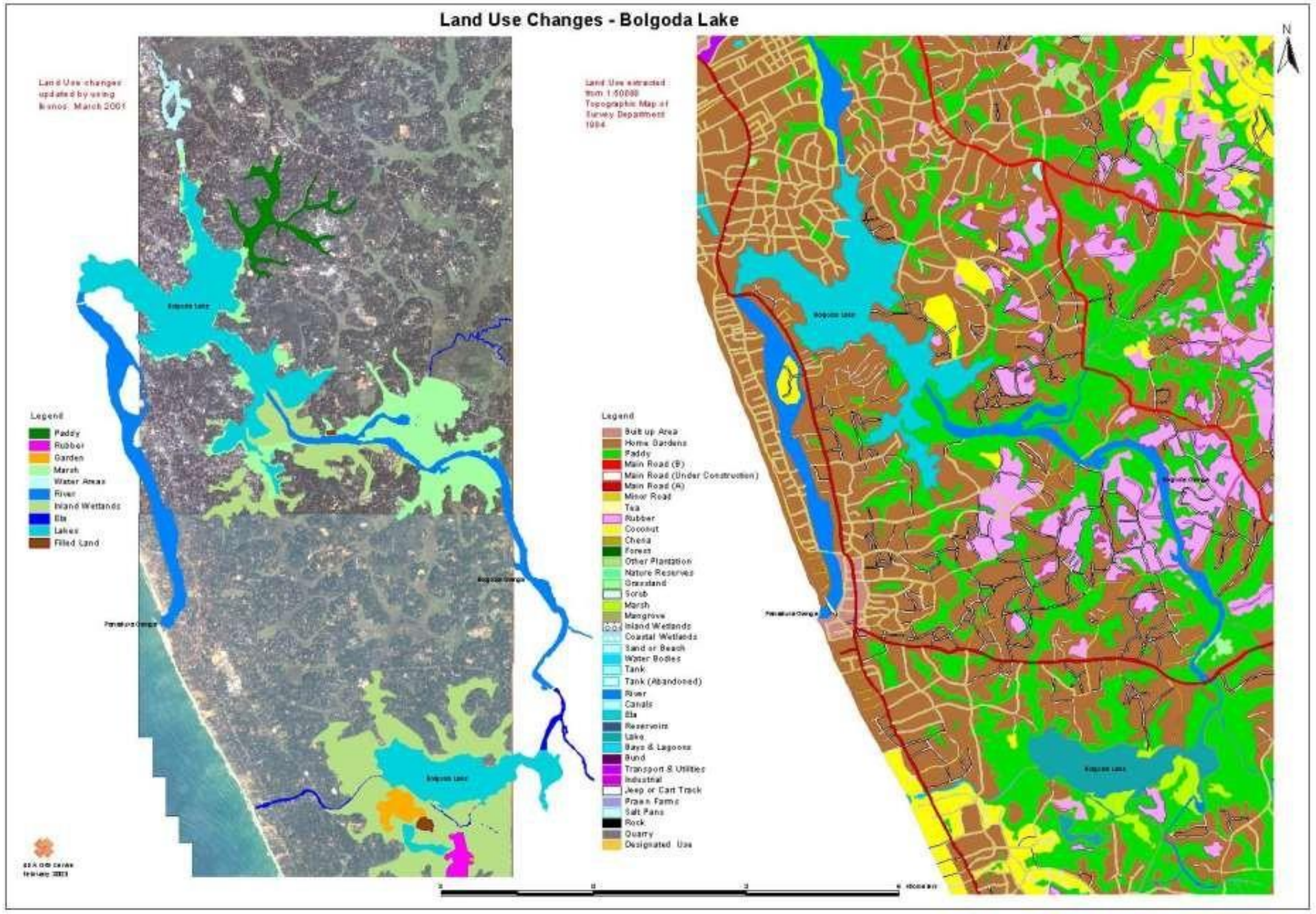




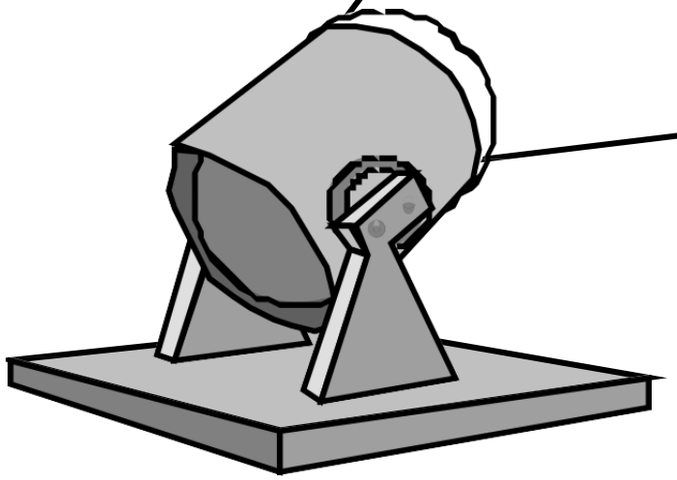

Roads in Colombo Municipal Council Area
(Based on IKONOS - 2005)


Ikonos Images
Veloor To Sampoor along the Proposed Road
N
Veloor
Nilaveli
Iqbalnagar
Periyakulam
Wilagama
Kanniya
Welveri
Kanniya
Muthithunagar
Muthithunagar
Iyawa North
Palampaddaru
Pokkuruni
Nadoothithu
Meeranagar
Soorangal
Upparu
Majeed Nagar
Iruthayapuram
Upparu
Peruveli
Nawarednapuram
Sampoor East
Naw
Kaddaiparichch
Chenaiyoor
LEGEND
Proposed Road
1 km Buffer
GND Boundary
UDA GIS Centre
June 2006

Monitoring Land Use changes



In short, a GIS does not hold maps or pictures -- it holds a database from which the data can be viewed in a new light.



Any Questions..???



